MIT 6.543
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Quantum Complexity Theory

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9/10/26

Lecture 1: Intro to BQP

- Complexity theory is about finding the mathematical limits of efficient computation

- Computers are physical objects, and efficiency is a physical notion (time, space, etc.)

So to understand complexity, need to use the right model of physics.. quantum mechanics

Mission #1 of quantum complexity theory:

Understand the power & limitations of quantum computers

– Not only are computers physical, many computational problems arise from physics as well.

Even natural processes can be thought of as computations.

Mission #2 of quantum complexity theory:

Use computation as a lens to understand nature!

The basic method of complexity theory is to classify problems into complexity classes typically defined by the computational resources needed to solve them

Recall $P = \{$ languages (= decision problems) solvable by poly-time classical algos $\}$

$BPP = \{$ " " " randomized classical algos w/ bounded error $\}$

if $x \in L$, $\Pr[M(x) = Yes] \geq 2/3$

$x \notin L$, $\Pr[M(x) = No] \leq 1/3$

} separated by a constant

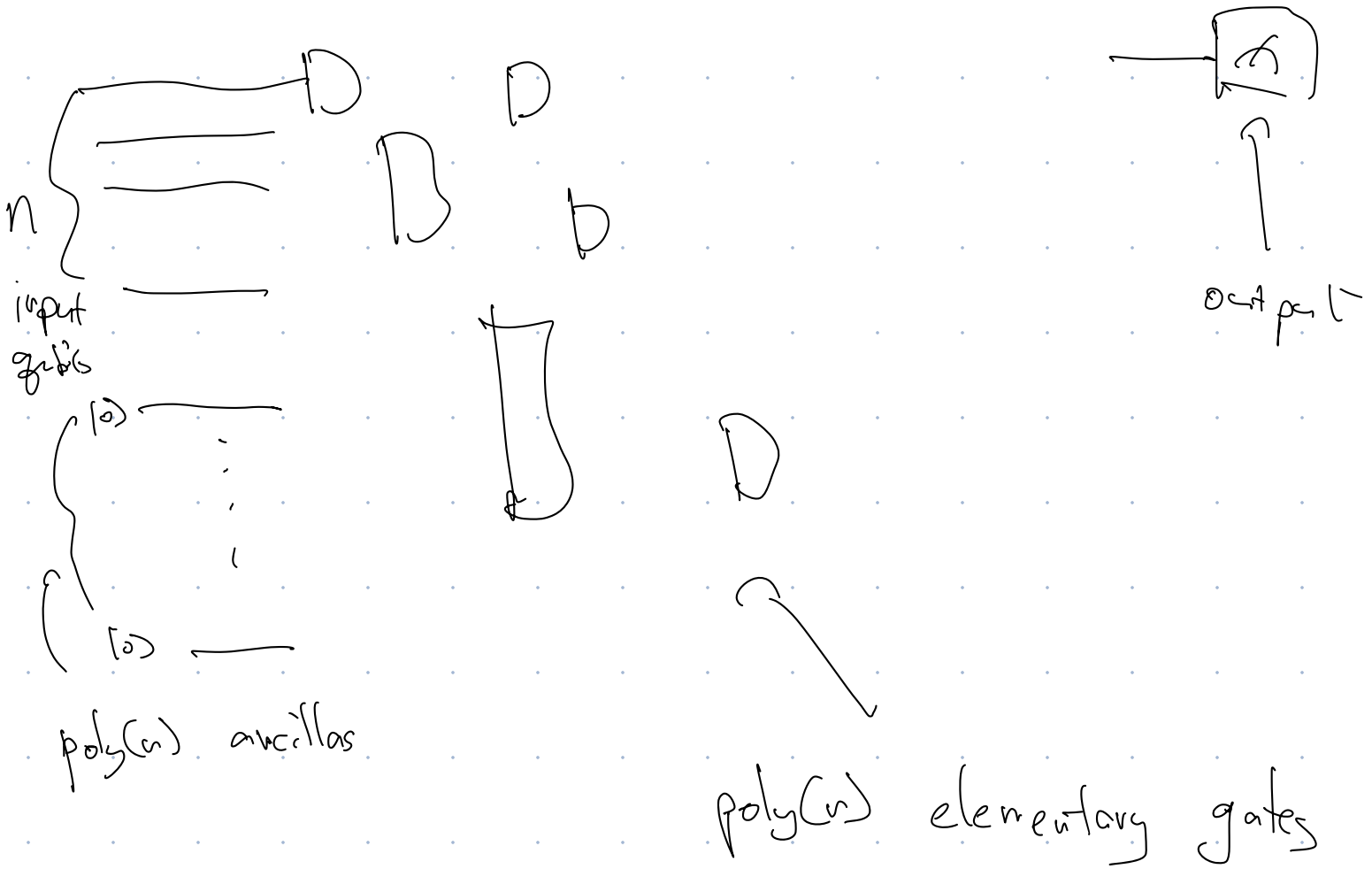
The star of quantum complexity theory is

$BQP = \{$ languages solvable by a poly-time b.e. quantum alg $\}$

what does this mean?

Classically, poly-time defined using Turing machines.

Quantumly, easier to use <u>circuits</u>



- No need to allow for intermediate measurements (just simulate coherently w/ ancilla)

- Can pick any reasonable universal set of elementary gates

  (SK says you only pay a polylog(n) cost to switch)

- Need some notion of uniformity: can't have a totally different circuit for each n

  "∃ classical TM M that runs in poly time and on input 1^n, spits out a description of the circuit for inputs of length n"

Easy to see

$$P \subseteq BPP \subseteq BQP$$

↑ by definition

↑ simulate random coins using Hadamards

- Also easy to see BQP is closed under complement (switching YES & NO)
- The thresholds $2/3$, $1/3$ are arbitrary and can be amplified

Just run the circuit many times and check that the fraction of accepts is above a threshold

Hoeffding's: if $X_1, \dots, X_k$ are indep. r.v. in $[0,1]$

$$\Pr\left[\left|\sum_{i=1}^{k} X_i - \mathbb{E}\sum_{i=1}^{k} X_i\right| \geq t\right) \leq 2e^{\frac{-2t^2}{k}}$$

So we can get $1 - \exp(-k)$ vs $\exp(-k)$ w/ k repetitions

One nice property of P & BPP is that they "respect subroutines": a P machine given access to an oracle deciding another P language is equiv to a P machine without the oracle

The same is true for BQP, but proving it requires some care

How does a q. circuit access an oracle? for L

$$U_L |x\rangle|b\rangle = \begin{cases} |x\rangle|b\rangle & \text{if } x \notin L \\ |x\rangle|b \oplus 1\rangle & \text{if } x \in L \end{cases}$$

?

unitary!

Suppose L is decided by a BQP machine.

The circuit gives some unitary U

but it is not this one!

$$U|x\rangle|0^{poly(n)}\rangle = \alpha|0\rangle|\varphi_0\rangle + \beta|1\rangle|\varphi_1\rangle$$

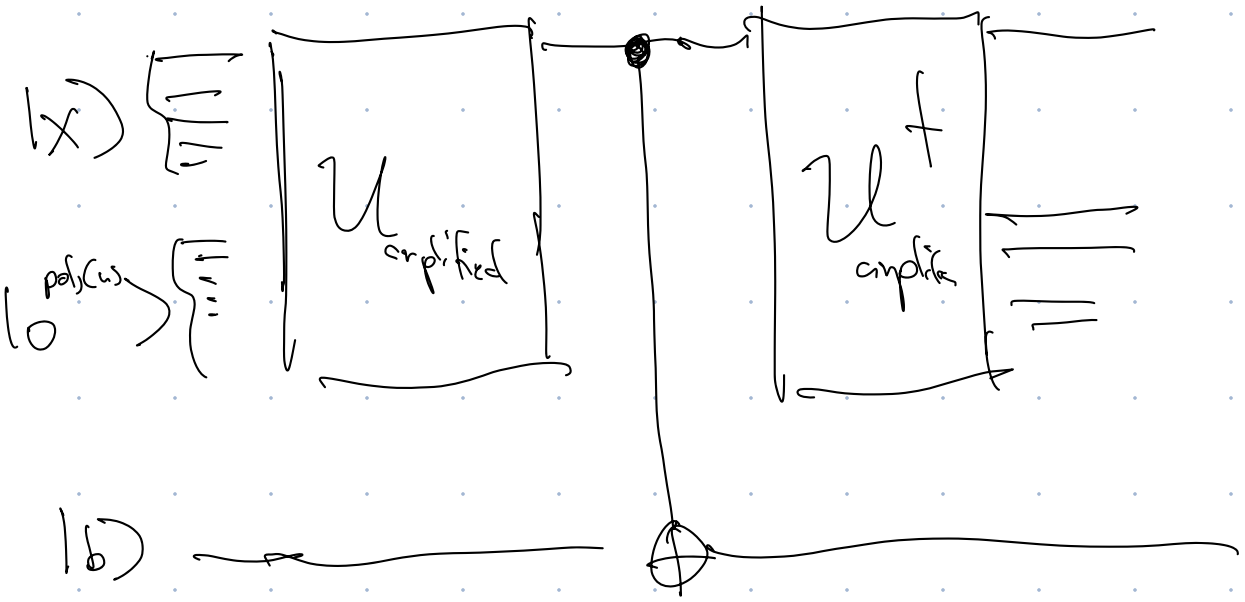
$|\beta|^2 \geq 2/3$ if $x \in L$

$\leq 1/3$ if $x \notin L$

can amplify to be exp close to 0 & 1

The real issue is the "junk" $|\varphi_0\rangle, |\varphi_1\rangle$

Solution: "uncompute"



This implements U_L as desired (except that we need ancilla, and there's some tiny error from amplification)

We have shown that

BQP

BQP = BQP

What about classical upper bounds on BQP?

$BQP \overset{??}{\subseteq} BPP$ We hope not!!

$BQP \overset{??}{\subseteq} NP$ Probably not, stay tuned for evidence

$BQP \subseteq EXP$ ✓

Prove this w/ brute force simulation:

can calculate $\left| (|x\rangle\langle 0| \otimes I)\, U |x\rangle|0\rangle \right|^2$

in exp time

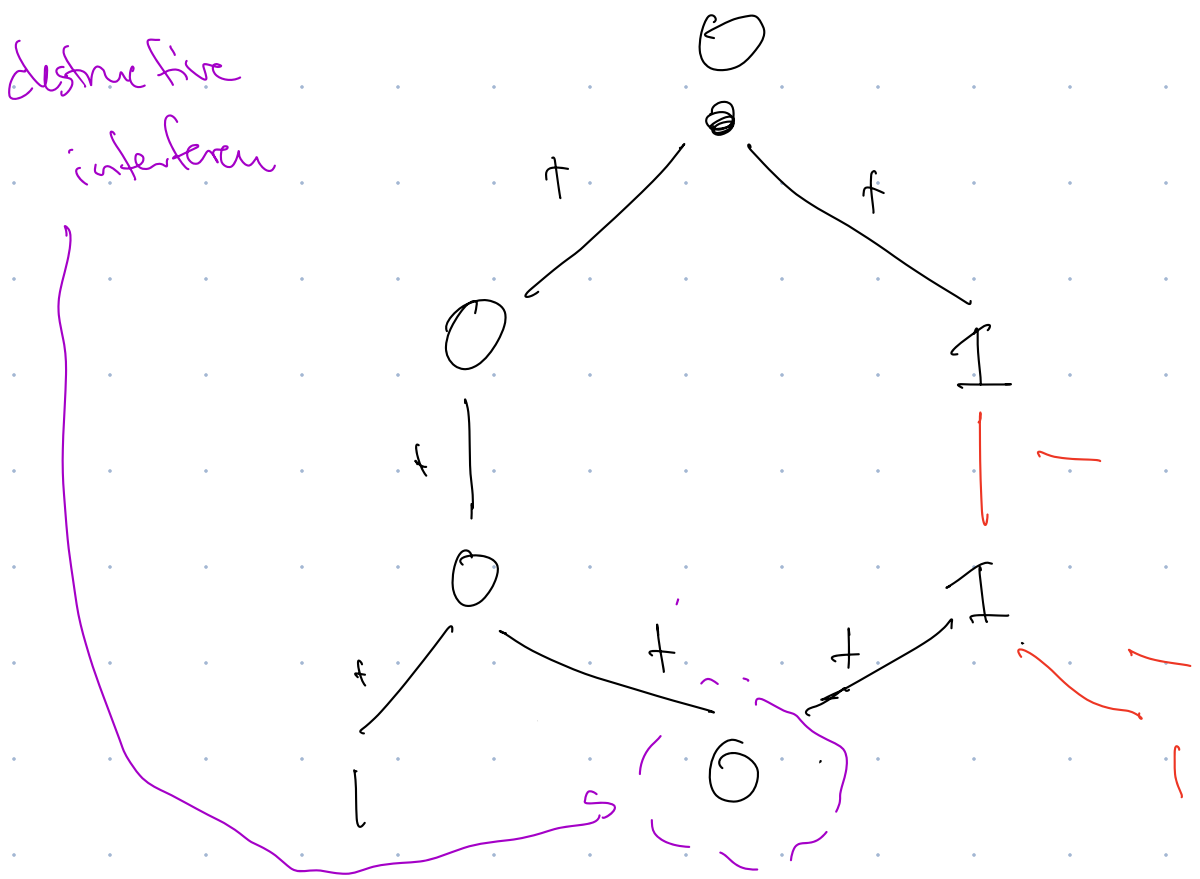
exp. big matrices and vectors

BQP ⊆ PSPACE

To show this, we need a new perspective: the "path integral"

Easiest to fix $\{H, CCZ\}$ as our universal set

destructive interference



In general, the state of the machine after a circuit with k Hadamard gates is

$\in \{\pm 1\}$

$$|\psi\rangle = \frac{1}{\sqrt{2^k}} \sum_{u \in \{0,1\}^n} \left(\sum_{\substack{\text{paths } p: \\ \text{end}(p) = u}} \text{sign}(p) \right) |u\rangle$$

can compute in poly time

$$\Pr[\text{accept}] = \frac{1}{2^k} \sum_{u \in \{0,1\}^{n-1}} \sum_{\substack{p, p': \\ \text{end}(p) = \text{end}(p') \\ = u}} \text{sign}(p)\,\text{sign}(p')$$

This is a big sum so it can be computed in PSPACE

$$BQP \subseteq PP$$

PP is like BPP without bounded error

if $x \in L$, $\Pr[\text{accept}] \geq 1/2$

if $x \notin L$, $\Pr[\text{accept}] < 1/2$

} no gap!

To show this, we use the path integral

$$\Pr[\text{accept}] = \frac{1}{2^k} \sum_{p, p' : \text{end}(p) = \text{end}(p')} \text{sign}(p)\,\text{sign}(p')\,(-1)^{\text{end}(p)_I}$$

$\Pr[\text{accept}] - \Pr[\text{reject}]$

β

$(-1)^{\text{end}(p)_I}$ ↑ output bit

PP machine needs to guess whether

β is $+$ or $-$.

Idea: just sample a random term in the sum!

PP alg:

- Sample p, p' uniformly at random
- If $\mathrm{end}(p) = \mathrm{end}(p')$:

  compute $S = \mathrm{sign}(p)\,\mathrm{sign}(p')\,(-1)^{\mathrm{end}(p)}$,

  if $S = +1$, output YES,

  else output NO.

Else:

output YES or NO uniformly at random.

$$\Pr[\text{accept}] = \Pr[\text{end}(p) \neq \text{end}(p')] \cdot$$

$$- \Pr[\text{reject}]$$

$$+ \Pr[\text{end}(p) = \text{end}(p')] \cdot$$ (prop. to. β)

Essentially this same calculation gives the tightest known classical upper bound on BQP

$$\text{BQP} \subseteq \text{AWPP}$$

↑ ugly def, look it up!

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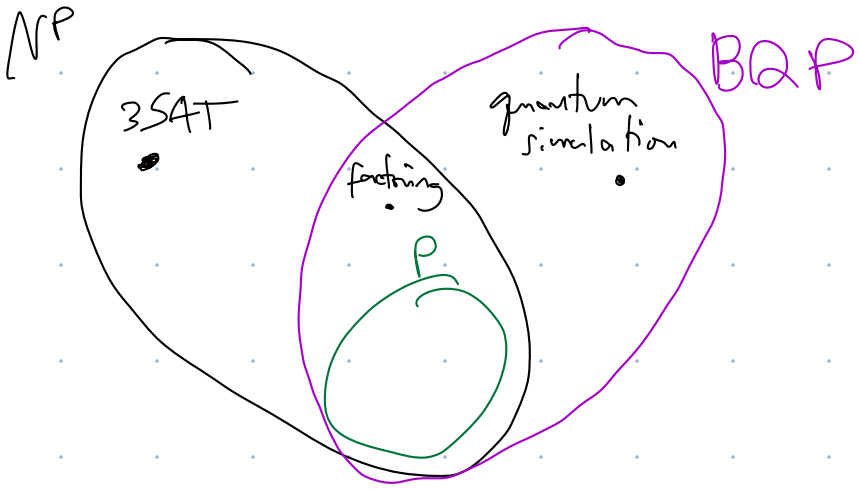
Lecture 2: BQP and NP, and the BBBV lower bound for unstructured search

Understanding the relationship between BQP & NP is of great importance

- If $BQP \stackrel{?}{\subseteq} NP$, then every quantum computation has an efficiently checkable certificate
- If $NP \stackrel{?}{\subseteq} BQP$, then all NP-complete problems (e.g. 3SAT, 3 coloring, TSP) can be solved by quantum computers!

We currently believe that neither is true.



Today we'll focus on evidence that $NP \not\subseteq BQP$.

What we will show is an oracle separation:

$\exists$ oracle O s.t.

$$NP^O \not\subseteq BQP^O$$

or unpacking this, there is some language L s.t. $L \in NP^O$ but $L \notin BQP^O$

L: related to O

$\notin$: provably

We can hope to prove this because deciding L requires some information about entries of O. So we will reduce the problem to the setting of "query complexity".

Note: An oracle separation does not imply that the classes are separated in reality!

$\exists$ oracles s.t. $P^O \neq NP^O$ and also

oracles s.t. $P^O = NP^O$

What it means is that any proof that settles P vs NP (or BQP vs NP) must be "non-relativizing": it must use techniques that break down in the presence of oracles.

Query complexity of search:

An oracle is a "magic box" that decides some language in unit cost.

Given input $x \in \{0,1\}^*$, O tells you "YES or NO"

Alternate perspective: The truth table of O is a computational input that can only be accessed by querying O on inputs $u \in \{0,1\}^*$

↑ index into truth table

This motivates the model of query complexity

- A "Query P" machine, given access to some input $X \in \{0,1\}^{2^n}$ (i.e. $N := 2^n$) should decide if $X \in S_{YES}^{(n)}$ or $X \in S_{NO}^{(n)}$ using $\text{poly}(n)$ queries to X ← only count queries, not runtime!
- A "Query BQP" machine is the same, but gets to make quantum queries

= apply $U_X : U_X|u\rangle|b\rangle \mapsto |u\rangle|b \oplus x_u\rangle$

A "Query NP" machine also gets a witness $w \in \{0,1\}^{poly(n)}$

- If $x \in S_{YES}^{(n)}$, $\exists$ w s.t. $M(w)$ outputs YES
- If $x \in S_{NO}^{(n)}$, $\forall w$, $M(w)$ outputs NO

It turns out that to find an oracle separation between BQP & NP, it will be enough to find a query promise problem specified by sets $S_{YES}^{(n)}$ & $S_{NO}^{(n)}$ for each n, s.t. a Query NP machine can solve it but a Query BQP machine cannot

Aside: a "promise problem" is a decision problem where the input is promised to come from some set. E.g. here, we only care about $x \in S_{YES}^{(n)} \cup S_{NO}^{(n)}$, not arbitrary $x \in \{0,1\}^n$

This "grey separation" can be lifted by very standard diagonalization arguments to an oracle separation $NP^O \not\subseteq BQP^O$, so we'll focus on the grey setting.

A candidate problem: Unstructured Search

NP is all about "finding a needle in a haystack." Think about SAT: want to find an input that makes the formula evaluate to 1, not 0.

Inspired by this, define the following problem

$$S_{YES}^{(n)} = \{ \text{all } x \in \{0,1\}^{2^n} \text{ except } 0^{2^n} \}$$

$$S_{NO}^{(n)} = \{ 0^{2^n} \}$$

Basically, want to decide if $\exists\, u \in \{0,1\}^n$ s.t. $x_u = 1$

& analogous to evaluating a SAT formula Φ at u

NP containment

Kind of obvious: a good witness is some w s.t. $\chi_w = 1$!

P lower bound:

Fix some Query P machine M.

If M solves Unstructured search, then

M must reject $x = 0^{2^n}$.

Let $u_1, u_2, \ldots, u_T$

↑ total # of queries

be the positions that M queries on

input 0^{2^n}. Since T is poly(n), it must be less than 2^n, so there is some location u' that has not been queried.

$x = 0\,0\,0 \ldots 0\,0\,0 \ldots 0\,0\,0 \ldots 0$

$u_1 \quad u_3 \quad u' \quad u_2 \quad u_T$

$x' = 0\,0\,0 \ldots 0\,1\,0 \ldots 0\,0\,0$

flip the value at u'

Now, $x' \in S_{YES}$, but M must still output NO on x', because x & x' look

identical to M!

Therefore, M gets x' wrong.

In fact, this argument applies to any deterministic machine that makes fewer than 2^n queries. Can even extend to randomized machines (see pset 1!)

BQP lower bounds

None of the ideas from the P case seem to work!

Problem 1: quantum algs. query in superposition. Even 1 query can "see" the entire input x

$$\frac{1}{\sqrt{2^n}} \sum_{u \in \{0,1\}^n} |u\rangle |0\rangle \xrightarrow{U_x} \frac{1}{\sqrt{2^n}} \sum_{u \in \{0,1\}^n} |u\rangle |x_u\rangle$$

We know this state can sometimes reveal a lot of information about x! E.g. the Bernstein Vazirani alg uses this to learn an entire linear function in one query.

Problem 2: specifically for unstructured search, Grover showed we can do way better than $\Omega(2^n)$ queries.

The # of queries needed is

$$O(\sqrt{2^n})$$

It turns out we can get around these problems and show that Grover's alg. is essentially optimal, by some more careful accounting.

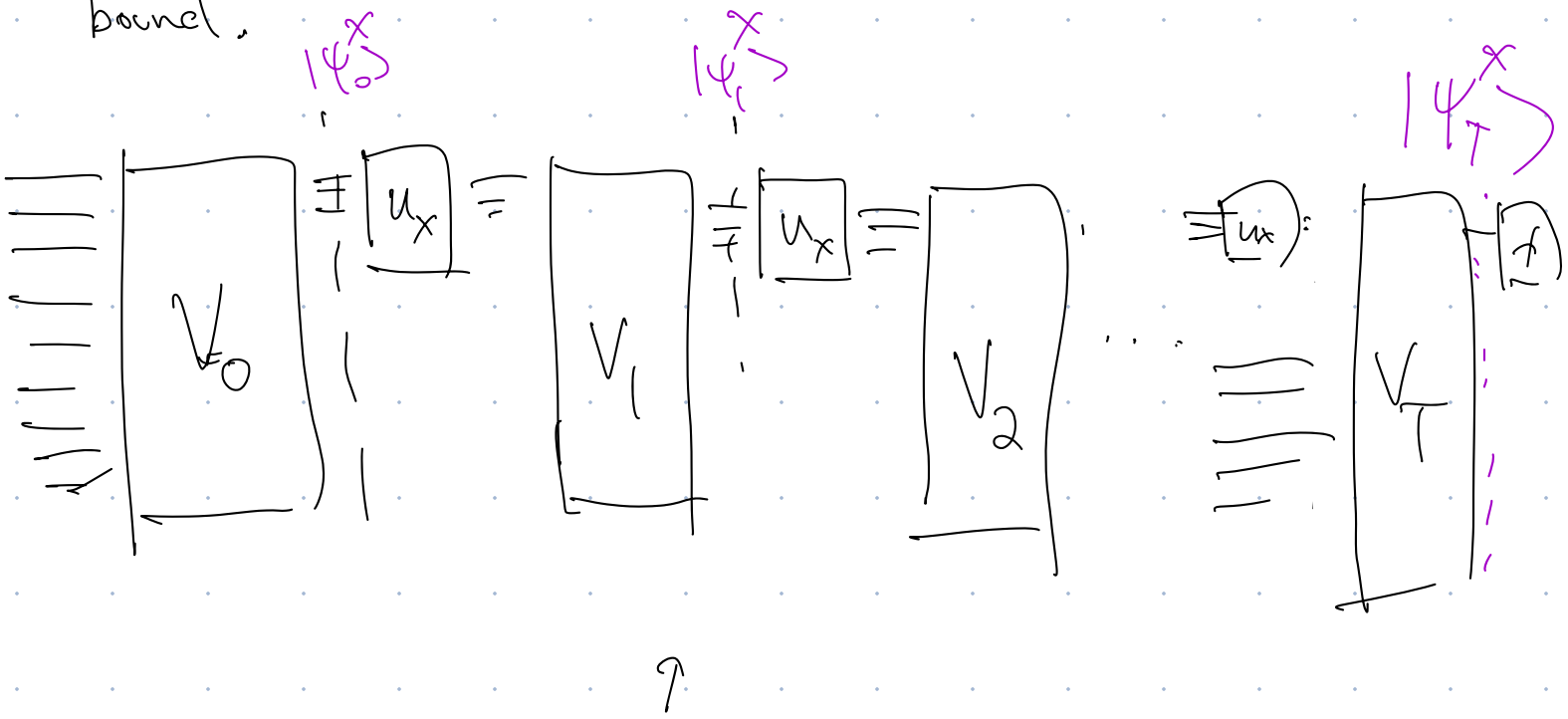
This is the

older than Grover's alg!

B '94 B B V

Bennett Bernstein Bassard Vazirani

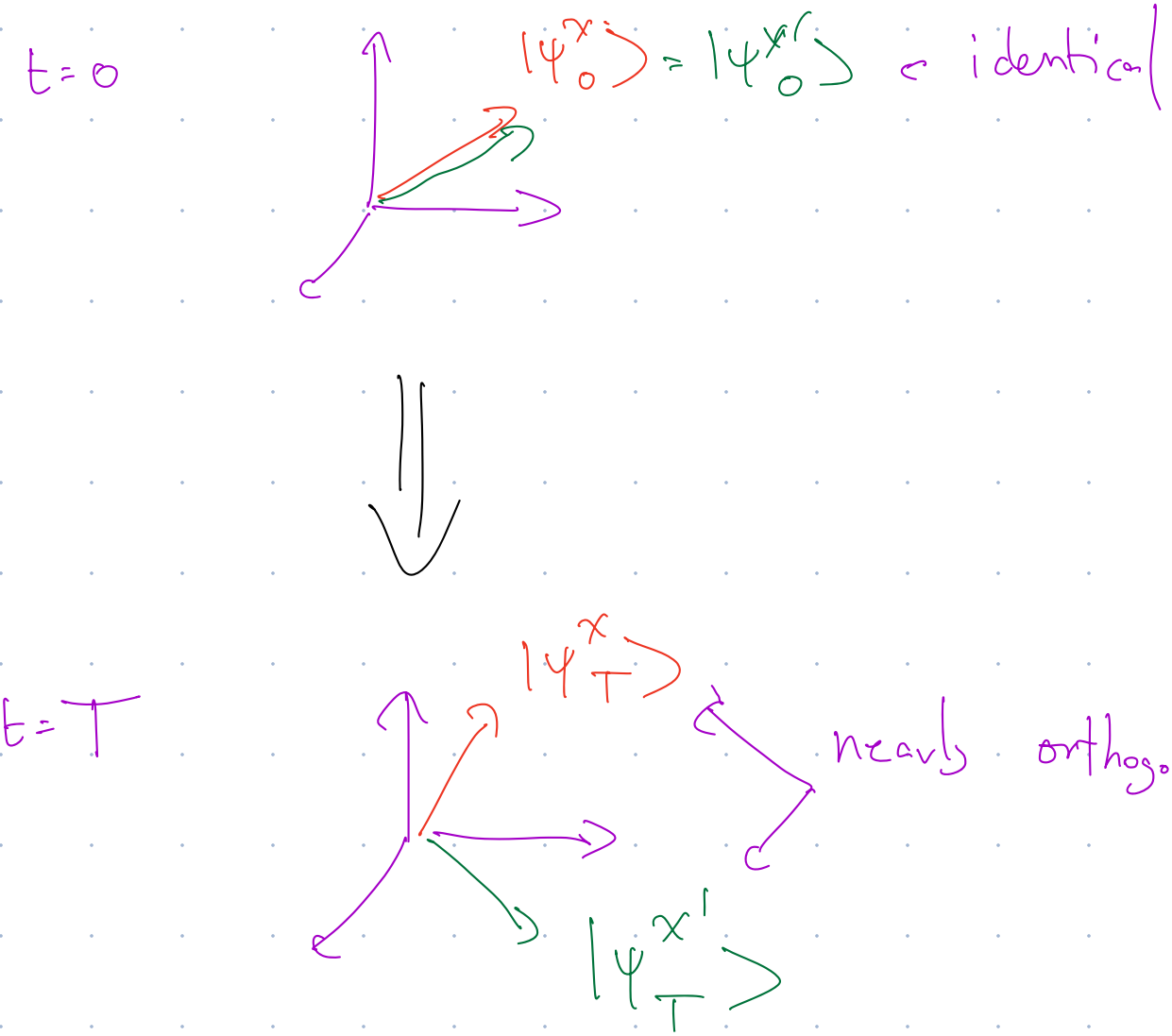
bound.



A general circuit making T queries.

Idea: For different x, x', track how far apart $|\psi_t^x\rangle$ and $|\psi_t^{x'}\rangle$ are

If $x \in S_{NO}$, $x' \in S_{YES}$, need



If a quantum alg. can pull apart the two states by only a little bit in Euclidean distance with each query, then T must be large for any successful quantum alg.

More precisely:

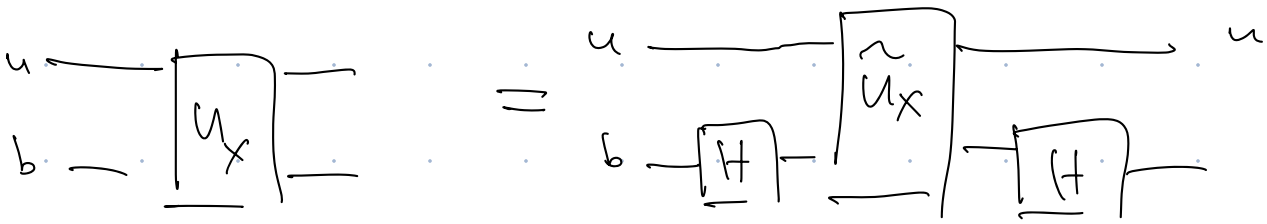
First, a slight change to the oracle:

instead of

$$U_x |u\rangle|b\rangle = |u\rangle|b + x_u\rangle,$$ work with

$$\tilde{U}_x |u\rangle|b\rangle = (-1)^{x_u \cdot b} |u\rangle|b\rangle$$

This is totally equivalent



Let $|\psi\rangle = \sum_{u,b} \alpha_{u,b} |u\rangle|b\rangle |\varphi_{u,b}\rangle$

oracle acts here only

We want to compute

$$\| \tilde{U}_x |\psi\rangle - \tilde{U}_{x'} |\psi\rangle \|_2$$

$$= \left\| \sum_{u,b} \alpha_{u,b} |u\rangle \left((-1)^{b \cdot x_u} - (-1)^{b \cdot x'_u} \right) |b\rangle |\varphi_{u,b}\rangle \right\|_2$$

$$= \left\| \sum_{u : x_u \neq x'_u} \pm 2 \alpha_{u,1} |u\rangle |1\rangle |\varphi_{u,1}\rangle \right\|_2$$

$$= 2 \sqrt{\sum_{u : x_u \neq x'_u} |\alpha_{u,1}|^2}$$

So to pull the states apart, the alg. must have high "superposition weight" on positions where x, x' differ.

We can now show our lower bound by a hybrid argument + induction

$$\Delta_t^{x,x'} :=$$

$$\| |\psi_t^x\rangle - |\psi_t^{x'}\rangle \|_2 = \| V_t \tilde{u}_x |\psi_{t-1}^x\rangle - V_t \tilde{u}_{x'} |\psi_{t-1}^{x'}\rangle \|_2$$

$$= \| \hat{u}_x |\psi_{t-1}^x\rangle - \hat{u}_{x'} |\psi_{t-1}^{x'}\rangle \|_2$$

$$= \| \hat{u}_x |\psi_{t-1}^x\rangle - \hat{u}_{x'} |\psi_{t-1}^x\rangle + \hat{u}_{x'} |\psi_{t-1}^x\rangle - \hat{u}_{x'} |\psi_{t-1}^{x'}\rangle \|_2$$

$$\leq \| (\hat{u}_x - \hat{u}_{x'}) |\psi_{t-1}^x\rangle \|_2 \quad := \eta_t^{x,x'}$$

triangle ineq.

$$+ \| \tilde{u}_{x'} (|\psi_{t-1}^x\rangle - |\psi_{t-1}^{x'}\rangle) \|_2$$

$$= \Delta_{t-1}^{x,x'}$$

Claim: Let $x = 0^N$, & let μ be the uniform distribution over weight-1 strings in $\{0,1\}^N$. Then

$$\mathop{\mathbb{E}}_{x' \sim \mu} \eta_t^{x,x'} \leq \frac{2}{\sqrt{N}} \quad \forall t$$

Pf: Let $|\psi_{t-1}^{x}\rangle = \sum \alpha_{u,b} |u\rangle |b\rangle |\varphi_{u,b}\rangle$

By our calculation above,

$$\mathop{\mathbb{E}}_{x' \sim \mu} \eta_t^{x,x'} = \mathop{\mathbb{E}}_{u \in [N]} 2 |\alpha_{u,I}|$$

↑ index where $x'_u = I$

$$= \frac{2 \sum_{u \in [N]} |\alpha_{u,I}|}{N}$$

$$= \frac{2}{N} \left\langle \vec{\alpha}, \overrightarrow{\mathrm{sgn}(\alpha)} \right\rangle$$

$$\begin{pmatrix} \alpha_{0,1} \\ \alpha_{1,1} \\ \alpha_{2,1} \\ \vdots \\ \alpha_{N,1} \end{pmatrix} \qquad \begin{pmatrix} \mathrm{sgn}(\alpha_{0,1}) \\ \mathrm{sgn}(\alpha_{1,1}) \\ \vdots \\ \mathrm{sgn}(\alpha_{N,1}) \end{pmatrix}$$

entries are ± 1

Cauchy-Schwarz

$$\leq \frac{2}{N} \cdot \|\vec{\alpha}\|_2 \cdot \|\overrightarrow{\mathrm{sgn}(\alpha)}\|_2$$

normalization of state

$$\leq \frac{2}{N} \cdot 1 \cdot \sqrt{N} = \frac{2}{\sqrt{N}}$$

Now, putting it all together, we have

$$\mathop{\mathbb{E}}_{x' \sim \mu} \Delta_T^{O^N, x'} \leq \mathop{\mathbb{E}}_{x' \sim \mu} \left(\eta_T^{O^N, x'} + \Delta_{T-1}^{O^N, x'} \right)$$

$$\leq \frac{2}{\sqrt{N}} + \mathop{\mathbb{E}}_{x' \sim \mu} \Delta_{T-1}^{O^N, x'}$$

$$\leq \frac{2}{\sqrt{N}} + \frac{2}{\sqrt{N}} + \mathop{\mathbb{E}}_{x' \sim \mu} \Delta_{T-2}^{O^N, x'}$$

$$\cdots$$

$$\leq \frac{2T}{\sqrt{N}} + \mathop{\mathbb{E}}_{x' \sim \mu} \Delta_0^{O^N, x'}$$

$$= 0$$

$\Rightarrow$ Any quantum alg. that solves unstructured search w/ bounded error needs $T = \Omega(\sqrt{N})$ queries!

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Lecture 3A: The Polynomial Method

The BBBV method is hard to directly apply to most problems. Sometimes alternate perspectives get us much farther by exploiting <u>symmetry</u> in the problem.

A very powerful one is the **polynomial method**, which views an algorithm as a low-degree polynomial.

$\equiv [V_0] = [\Theta] = [V_1] = [\Theta] = [V_2] \ldots$

<u>Lemma:</u> After t steps, the state $|\psi_t\rangle = \sum \alpha_i |i\rangle$ (α_i: degree-t poly in the bits $x_1 \ldots x_N$ of the oracle)

<u>Pf:</u> By induction!

Base case: $t=0$, obvious

$t \to t+1$

$$O|\psi_t\rangle = O \sum_{i,b,z} \alpha_{i,b,z} |i\rangle|b\rangle|z\rangle$$

$$= \sum_{i,b,z} (-1)^{b \cdot x_i} \alpha_{i,b,z} |i\rangle|b\rangle|z\rangle$$

$$= \sum_{i,b,z} (1 - 2bx_i)\, \alpha_{i,b,z} \quad |i\rangle|b\rangle|z\rangle$$

degree 1 · degree t

degree $t+1$ polynomial.

∎

Cor: $\Pr[\text{acceptance}^x] =$ degree $2T$ poly in $x_1 \dots x_N$ of T-query alg.

The 2 comes from squaring!

So if you can show that no low-degree poly matches the promise problem you want to decide, you show that there's no q. alg!

But it's very hard to reason about multivariate polynomials with lots of variables. So often one uses symmetry of some kind to reduce the number of variables.

The unstructured search problem:

Idea: consider distributions on inputs that are symmetric under permutations of x

$\mu_r :=$ uniform dist. over x s.t. $|x| = r$

(Hamming weight)

Claim: If $p(x_1, \dots, x_N)$ is degree d multivariate,

$q(r) := \mathop{\mathbb{E}}_{x \sim \mu_r} p(x_1, \dots, x_N)$ is degree d poly in r.

Pf: $\mathop{\mathbb{E}}_{x \sim \mu_r} x_1 x_3 x_{17} = \Pr_{x \sim \mu_r}[\text{all 3 of } x_1, x_3, x_{17} \text{ are equal to } 1]$

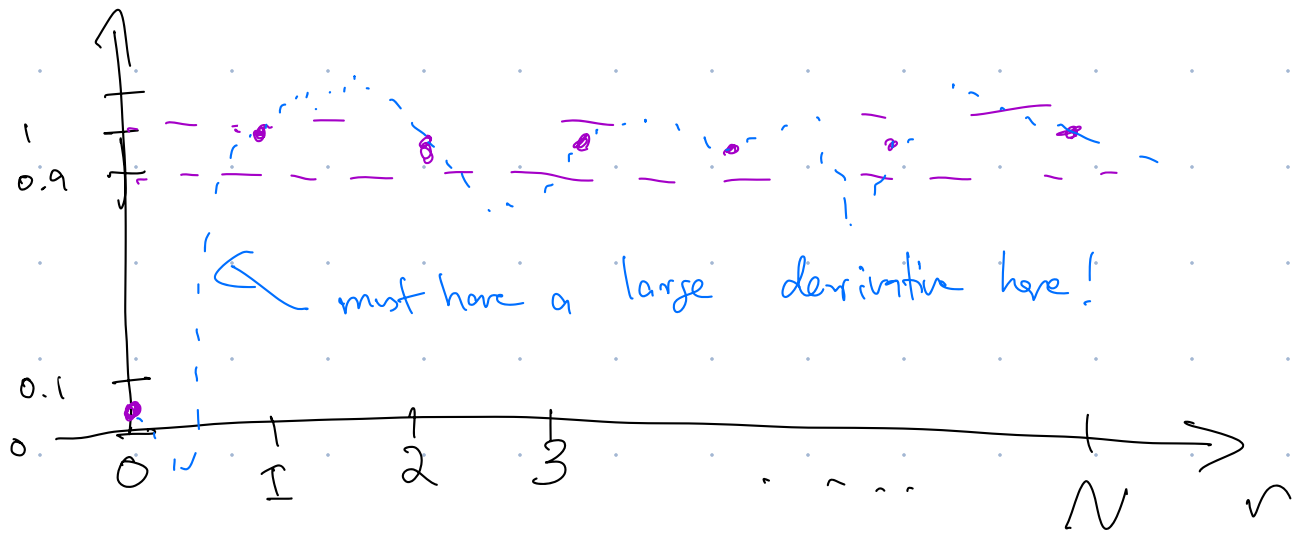
$$= \frac{\binom{N-3}{r-3}}{\binom{N}{r}}$$

$$= \frac{\frac{(N-3)!}{(r-3)!\,(N-r)!}}{\frac{N!}{r!\,(N-r)!}}$$

$$= \frac{r(r-1)(r-2)}{N(N-1)(N-2)}$$ ← degree 3 poly in r!

Easily generalizes to monomials of any degree.

Now, if we have an alg that solves unstructured search in T queries $\Rightarrow$ a degree-$2T$ polynomial q that looks like this



Note: no constraint on q outside of integer points (need not even be positive)

It turns out no low-degree poly can look like this!

Markov's other inequality:

$$\max_{-1 \le x \le 1} \left| \frac{dp(x)}{dx} \right| \le d^2 \max_{-1 \le x \le 1} |p(x)|$$

d^2: degree(p)

This immediately gives us a useful bound if q stays bounded at non-integer points.

If q gets large at non-integer points, then you get a large derivative in between integer points and you can still apply this.

Upshot: Need $\deg(q) \ge \Omega(\sqrt{N})$.

Getting the right symmetrization is a bit of an art!

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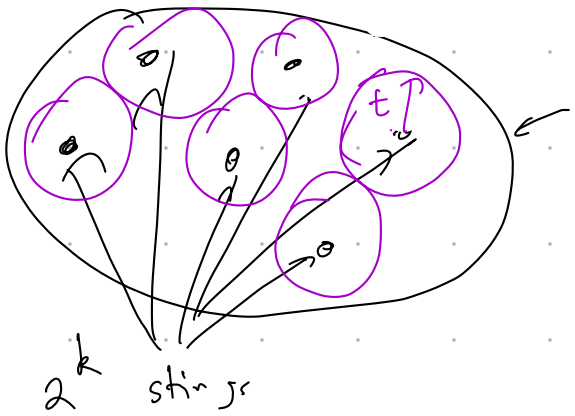
Lecture 4: Error Correction Bootcamp

A classical code is a collection of bit strings

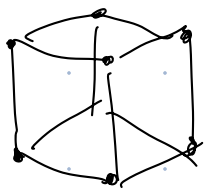


Can correct any error on at most t locations

Quantum codes:

Classical errors are discrete, but quantum errors are continuous

Solution: discretize error using linearity of Q.M.



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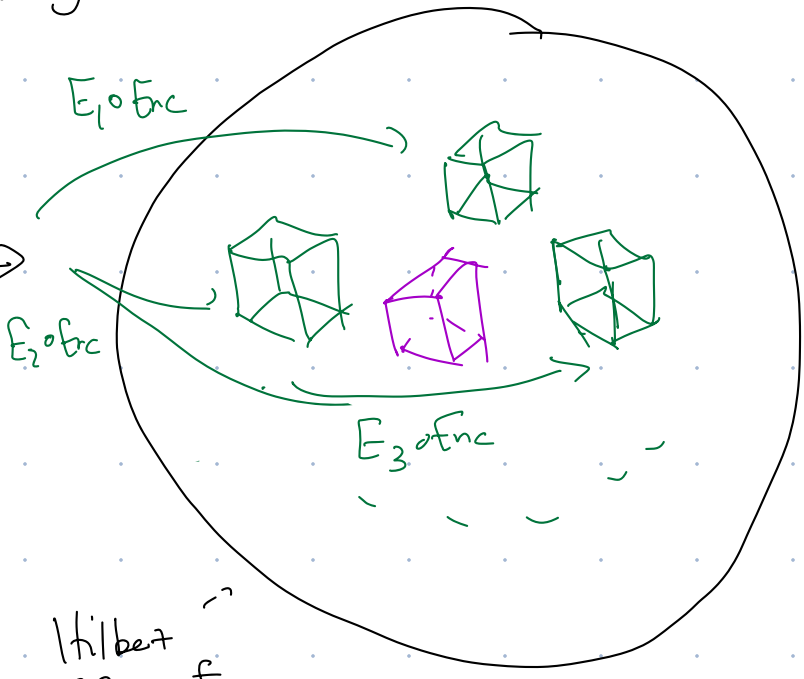
Hilbert space of dimension 2^k

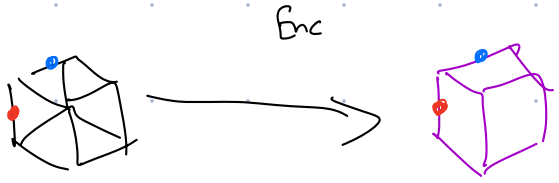
Enc



Hilbert space of dimension 2^n

For now, think of $E_1, E_2, \ldots$ as a basis for the set of errors we want to correct.

In order to recover information:



has to preserve the structure of the subspace

$$\langle a|b\rangle = 0 \iff \langle \bar{a}|\bar{b}\rangle = 0$$

$\langle \bar{a}| = (Enc|a\rangle)^\dagger$, $|\bar{b}\rangle = Enc|b\rangle$

Similarly, for each error E_i, E_j

$$\langle a|b\rangle = 0 \Rightarrow \langle \bar{a}|E_i^\dagger E_j|\bar{b}\rangle = 0$$

More generally, what does it mean to correct an error?

$\exists\ U_{rec}$ s.t. $U_{rec} E_i|\bar{\psi}\rangle = |\psi\rangle \otimes |junk\ i\rangle$

U_{rec}: "recovery map"

$|junk\ i\rangle = \frac{1}{\||E_i|\psi\rangle\|}|syndrome\ i\rangle$

By linearity, this implies that for all $|\psi\rangle$, $\|E_i|\bar{\psi}\rangle\|$ is the same value $\sqrt{p_{E_i}}$ depending only on i.

Now we can derive some more conditions:

$$\langle \bar{a} | E_i^\dagger E_j | \bar{b} \rangle = \langle \bar{a} | E_i^\dagger U_{rec}^\dagger U_{rec} E_j | \bar{b} \rangle$$

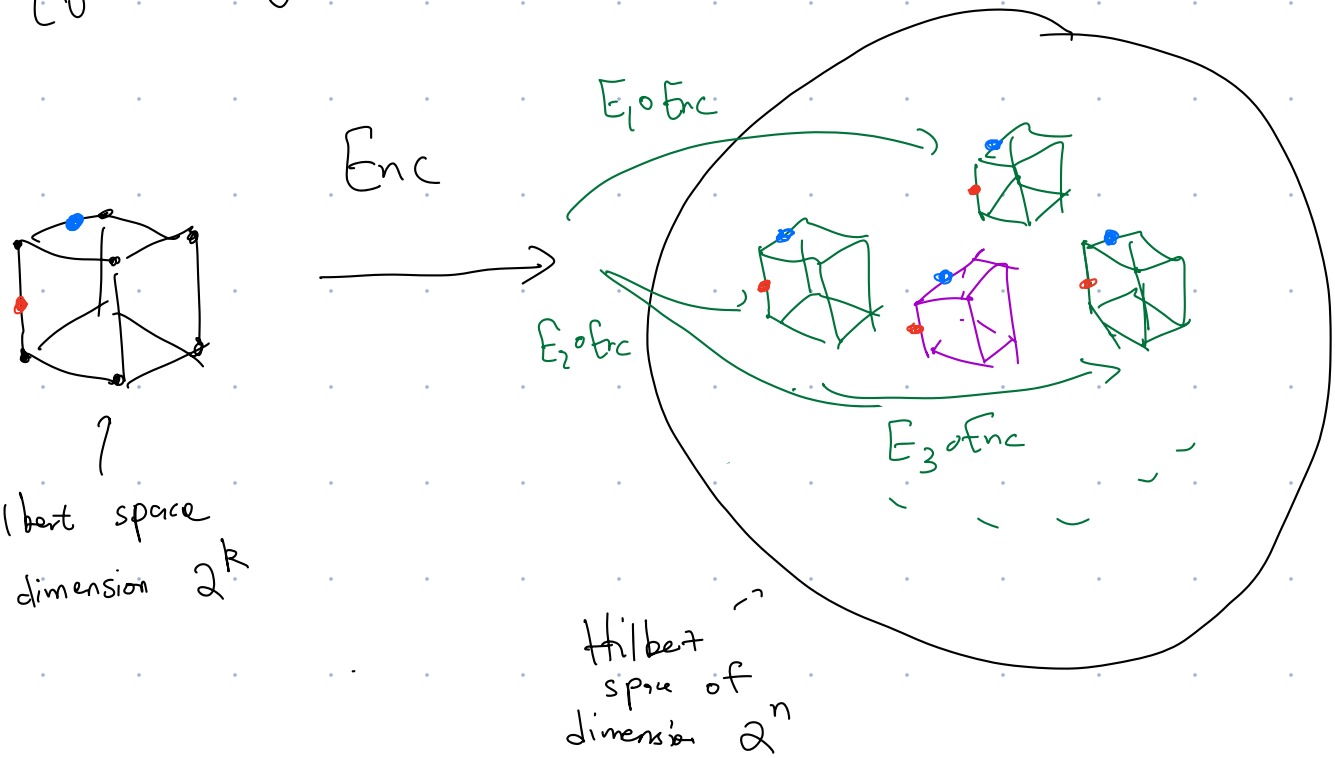
$$= \langle a | b \rangle \cdot \frac{\langle \text{syndrome}_i | \text{syndrome}_j \rangle}{\sqrt{p_{E_i} \cdot p_{E_j}}}$$

$$= \langle a | b \rangle \cdot \underbrace{\eta_{ij}}_{\text{independent of } |a\rangle, |b\rangle} \qquad (*)$$

Equation (*) is called the Knill-Laflamme condition and turns out to be sufficient for U_{rec} to exist! (We won't prove this here)

A useful special case for intuition is when

$$\eta_{ij} = \delta_{ij}.$$



In this special case, each "box" in the encoded space is an orthogonal "copy" of the original 2^k-dimensional Hilbert space.

U_{rec} operates by "extracting" the information of which box the state is in, without disturbing the state itself.

Pauli Errors:

Typically, we take $\{E_1, E_2, \dots\}$ to be the set of all Pauli operators supported on up to weight t. In this case, we say the code corrects "all weight t errors"

Classical vs. Quantum Codes:

Intuition: classical codes operate by copying information.

E.g. the repetition code

$$0 \to \overline{0} = 000\dots0$$
$$1 \to \overline{1} = 111\dots1$$

Quantum codes cannot work like this, b/c of the no-cloning thm.

Indeed, quantum codes satisfy the local indistinguishability property: if a code corrects t errors, then all code states look the same on regions of size $\leq t$.

Pf: Let $|a\rangle$, $|b\rangle$ be two states

Enc $\downarrow$ $\downarrow$

$|\bar{a}\rangle$ $|\bar{b}\rangle$.

To show that $|\bar{a}\rangle, |\bar{b}\rangle$ are locally indistinguishable, let M be a measurement operator supported on a region T of $\leq t$ qubits. Then we must show that $\langle \bar{a}| M |\bar{a}\rangle = \langle \bar{b}| M |\bar{b}\rangle$

Indeed,

$$\langle \bar{a}| M |\bar{a}\rangle = \langle \bar{a}| \sum_i m_i E_i |\bar{a}\rangle$$

$$= \sum_i m_i \langle \bar{a}| E_i |\bar{a}\rangle$$

$$= \sum_i m_i \langle \bar{a}|\bar{a}\rangle \cdot \eta_{0i} \quad \hookleftarrow \text{KL}$$

$$= \sum_i m_i \langle \bar{b}|\bar{b}\rangle \eta_{0i}$$

$$= \sum_i m_i \langle \bar{b}| E_i |\bar{b}\rangle$$

$$= \langle \bar{b}| M |\bar{b}\rangle$$

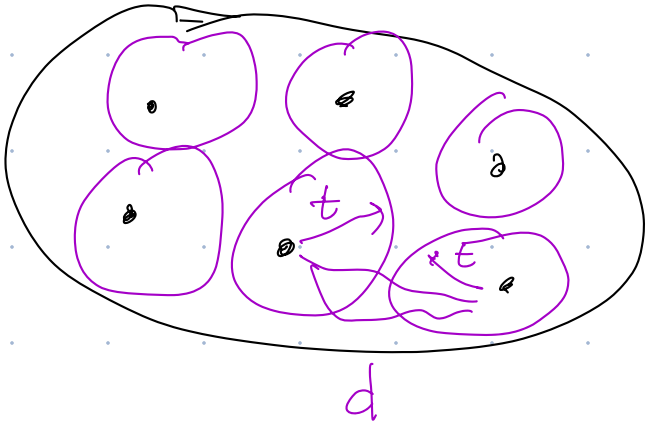
Intuition: quantum codes "stretch out" information using entanglement

Constructions: linear & stabilizer codes

Classically: a linear code is a subspace of $\mathbb{F}_2^n$ of dimension $k \Leftarrow 2^k$ vectors

Distance := min Hamming weight of a nonzero codeword
= min distance between codewords
(by linearity, difference between codewords is itself a codeword)

A code with distance d can correct $\left\lfloor \frac{d-1}{2} \right\rfloor$ errors



Notation: $[n, k, d]$ code

Examples: 1) Repetition code $[n, 1, n]$

$\text{span}\{00\ldots0, 11\ldots1\}$

2) Reed-Solomon code

Details are a little complex but here's the intuition.

Let $\mathbb{F}_q$ be a finite field of order q

$|\mathbb{F}_q| = q$ ← represent each element with $\log q$ bits

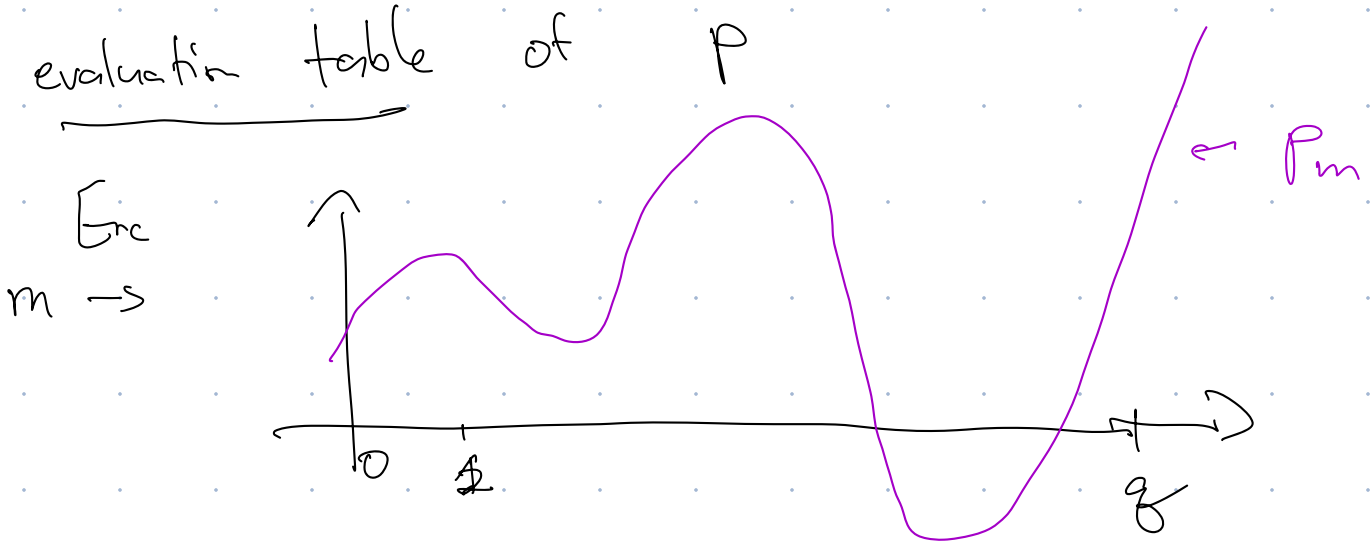
A finite field is a finite set that supports addition, multiplication, & division. So we can think of polynomials over it.

Given a message $m \in \mathbb{F}_q^k$, encode it into a polynomial

$$P_m(x) = m_1 + m_2 x + m_3 x^2 + \cdots + m_k x^{k-1}$$

degree $k-1$

Then the encoding of m will be the evaluation table of P



Fact: Any two degree-r polys that agree on $r+1$ points are actually identical!

$\Rightarrow$ If $m \neq m'$, $Enc(m)$ & $Enc(m')$ differ on at least $q-k+1$ points

So we have a

$$[\,q,\ k,\ q-k+1\,]_q$$

code over $\mathbb{F}_q$ (can convert to bits if needed)

Quantumly: The analog of a linear code is a stabilizer code.

In a classical linear code, the code subspace S is the null space of some matrix H ← "parity check matrix"

$$\begin{pmatrix} -h_1- \\ \vdots \\ -h_e- \end{pmatrix}$$

$$x \in S \iff \langle h_i, x \rangle = 0 \quad i \in [e]$$

In a stabilizer code, the code subspace is the common +1 eigenspace of a collection of Pauli operators, called stabilizer generators

$$|\psi\rangle \in S \iff \langle\psi| P_i |\psi\rangle = I \quad i \in [e]$$

Generalizes parity checks

$h_i \to Z^{h_i}$

Very often, can get away with taking the stabilizer generators to be either tensor products of X's or tensor products of Z's.

Such codes are called CSS

CSS code = a classical linear code in the X basis and Z basis

9/24/26

Lecture 5: The Yamakawa-Zhandry Problem and the role of structure in quantum speedups

When defining query complexity, we introduced the concept of promise problems: only need to solve the problem on $x \in S_{YES} \cup S_{NO} \subsetneq \{0,1\}^N$.
possibly

All the famous examples of exponential quantum speedup are for problems where the promise set is highly structured:

- Bernstein-Vazirani
- Simon's
- Welded trees

And even in the non-oracle world, Shor's alg solves a highly structured task.

Question: Is structure necessary for quantum speedups?

A lot of interesting results around this in the query setting.

Thm: [Beals et al '98]

For any total function $S_{YES} \cup S_{NO} = \{0,1\}^N$, the quantum query complexity is polynomially related to the classical deterministic query complexity.

deterministic → $D(f) = O(Q(f)^6)$ ← quantum

(Shown via polynomial method!)

So some sort of promise is necessary for exponential speedup. In fact, it's conjectured that a classical machine can efficiently simulate a quantum one on any constant fraction of $\{0,1\}^N$.

Conjecture [Aaronson-Ambainis '09]:

Let Q be a quantum alg. making T queries to a random oracle $\mathcal{O}$. Then $\forall\ \varepsilon, \delta > 0$, $\exists$ a classical determinist alg. C making $\text{poly}(T, 1/\varepsilon, 1/\delta)$ queries s.t. on $1-\delta$ fraction of oracles $\mathcal{O}$,

$$C^O \approx_\varepsilon \Pr[Q^O = \text{accept}]$$

In other words, any quantum grey alg. can be classically simulated on most of its inputs. This conjecture is still widely believed and progress has recently been made towards solving it...

But, the AA conjecture is only for decision problems!

Yamakawa-Zhandry '22: There exists a search problem where quantum achieves an exponential speedup w.h.p. for random oracles. Moreover, this problem is in NP! (Can efficiently check soln.)

So even relatively unstructured problems admit exponential q. speedups!

YZ is inspired by ideas from lattice-based cryptography, and in turn has inspired the DQI algorithm for combinatorial optimization.

The YZ problem:

Let Σ be some big alphabet, and let C be a (linear) code over Σ^n.

Think Flag

Let $h_i : \Sigma \to \{0,1\}$ be a function for each $i \in [n]$.

"hash function"

encoded in the oracle

The task is to find $x \in C$ s.t.

$$(h_1(x_1), h_2(x_2), \ldots, h_n(x_n)) = (0, 0, \ldots, 0)$$

This is clearly in FNP ("Search NP") for all values of h:

given an x, just check whether it's in C, and that each coordinate hashes to I.

Thm: There is a quantum alg making poly(n) queries that solves this problem for a random h with very high probability. OTOH, no classical alg making subexponentially many queries can do this. (for a certain choice of code!)

Classical hardness:

The classical hardness will use a property called "list-recoverability". Essentially, this means that few codewords are consistent with the locations in h queried by the algorithm.

Def: A code $C \subseteq \Sigma^n$ is (ζ, ℓ, L)-list recoverable if for any subsets $S_i \subseteq \Sigma$, $|S_i| \leq \ell$ for each i.

$$\left| \{ (x_1, \ldots, x_n) \in C : |\{i : x_i \in S_i\}| \geq (1-\zeta)n \} \right| \leq L$$

"At most L codewords are consistent with a $1-\zeta$ fraction of the lists, if each list is of size at most ℓ"

Code parameters: Our code will be (ζ, ℓ, L)-list rec. for $\zeta = \Omega(1)$, $\ell = 2^{\alpha n^c}$, $L = 2^{\tilde{O}(n^{c'})}$

$0 < c < 1$ $\quad$ $0 < c < c' < 1$

Idea: Suppose a classical alg. makes less than ℓ queries to h. Then there are at most L codewords consistent with the queries.

The chance that any fixed $x \in C$ is a solution is

$$\Pr_h \left[h_1(x_1) = 0 \wedge h_2(x_2) = 0 \wedge \dots \right] = \frac{1}{2^n}$$

So by union bound, nothing in the list is a solution.

This is just an intuitive sketch, ignoring adaptivity of queries, but "morally" this is the proof!

Quantum easiness:

This is the fun part: we will see a very clever algorithmic trick!

Dream: Observe that it's easy to prepare the following states:

$$|\psi_C\rangle = \frac{1}{\sqrt{|C|}} \sum_{x \in C} |x_1 x_2 \dots x_n\rangle$$

$$|\psi_h\rangle = \frac{1}{\sqrt{|h^{-1}(0)|^n}} \sum_{\substack{x \in \Sigma^n : \\ h_1(x_1)=0 \\ \vdots \\ h_n(x_n)=0}} |x_1 x_2 \dots x_n\rangle$$

We want a point in the intersection of the supports of these two states. Suppose we could compute the state

$$|\varphi\rangle \propto |\psi\rangle \odot |\psi_n\rangle$$

$\odot$: entrywise product of coefficients.

Then measuring $|\varphi\rangle$ will give us a point in the intersection!

It will turn out that if the code has some extra properties, then this multiplication is possible!

Entrywise multiplication:

Let's try going into the Fourier domain!

Recall the FT over prime finite fields

$$\mathcal{F}|x\rangle = \frac{1}{\sqrt{q}} \sum_y \omega^{x \cdot y} |y\rangle$$

ω: q'th root of unity

A Fourier transform converts products into convolutions:

$$|\alpha\rangle = \sum_x \alpha_x |x\rangle \qquad |\beta\rangle = \sum_y \beta_y |y\rangle$$

$$\mathcal{F}|\alpha\rangle = \sum_x \hat{\alpha}_x |x\rangle \qquad \mathcal{F}|\beta\rangle = \sum_y \hat{\beta}_y |y\rangle$$

$$= \frac{1}{\sqrt{q}} \sum_z \omega^{xz} \alpha_z \qquad = \frac{1}{\sqrt{q}} \sum_z \omega^{yz} \beta_z$$

Inverse

$$\alpha_x = \frac{1}{\sqrt{q}} \sum_z \omega^{-xz} \hat{\alpha}_z$$

$$|\alpha\rangle \odot |\beta\rangle = \sum_x \alpha_x \beta_x |x\rangle$$

$$\mathcal{F}|\alpha\rangle \odot |\beta\rangle = \frac{1}{\sqrt{q}} \sum_{xy} \omega^{xy} \alpha_x \beta_x |y\rangle$$

$$= \frac{1}{\sqrt{q}^3} \sum_y \left(\sum_{xuv} \omega^{xy - xu - xv} \hat{\alpha}_u \hat{\alpha}_v \right) |y\rangle$$

Sum over x vanishes unless

$$y - u - v = 0 \iff v = y - u$$

$$= \frac{1}{\sqrt{q}} \sum_y \sum_u \hat{\alpha}_u \hat{\alpha}_{y-u} |y\rangle$$

convolution

$$= \frac{1}{\sqrt{q}} \sum_y \sum_u \hat{\alpha}_u \hat{\alpha}_y |y+u\rangle$$

↖ change of variables

normalization

$$|\alpha\rangle|\beta\rangle \xrightarrow{F\otimes F} |\hat{\alpha}\rangle|\hat{\beta}\rangle = \sum_{u,v} \hat{\alpha}_u \hat{\beta}_v |u\rangle|v\rangle$$

coherent addition

$$\longmapsto \sum_{u,v} \hat{\alpha}_u \hat{\beta}_v |u\rangle|v\rangle|u+v\rangle$$

uncompute

$$\longmapsto \sum_{u,v} \hat{\alpha}_u \hat{\beta}_v |0\rangle|v\rangle|u+v\rangle$$

???

$$\longmapsto \sum_{u,v} \hat{\alpha}_u \hat{\beta}_v |0\rangle|0\rangle|u+v\rangle$$

Need to coherently uncompute v from $u+v$!

Idea: suppose the states $|\alpha\rangle, |\beta\rangle$ are structured so that u is a codeword of some code, and v is low weight (below distance).

Then $|u+v\rangle|v\rangle \longrightarrow |u+v\rangle|0\rangle$ is the inverse of the decoding unitary!

For us:

$$|\alpha\rangle = |\psi_C\rangle = \frac{1}{\sqrt{|C|}} \sum_{x \in C} |x\rangle$$

$$|\hat{\alpha}\rangle = \frac{1}{\sqrt{|C|}} \cdot \frac{1}{\sqrt{q^n}} \sum_{\substack{x \in C \\ z \in \mathbb{F}^n}} \omega^{z \cdot x} |z\rangle$$

This sum vanishes unless $z \in C^\perp$

$$= \sqrt{\frac{|C|}{q^n}} \sum_{z \in C^\perp} |z\rangle = \frac{1}{\sqrt{|C^\perp|}} \sum_{z \in C^\perp} |z\rangle$$

For a linear code C, $C^\perp$ is called the dual code

$$|\beta\rangle = |\psi_h\rangle = \left(|h_1^{-1}(0)\rangle \right) \otimes \left(|h_2^{-1}(0)\rangle \right) \otimes \cdots$$

where each $$|h_i^{-1}(0)\rangle = \frac{1}{\sqrt{|h_i^{-1}(0)|}} \sum_{\substack{x \in \Sigma: \\ h_i(x) = 0}} |x\rangle$$

$|h_i^{-1}(0)| \approx \frac{q}{2}$

$$|h_i^{-1}(0)\rangle \approx \sqrt{\frac{2}{q}} \sum_{x \in \Sigma} \mathbb{I}[h_i(x)=0]|x\rangle$$

$$= \sqrt{\frac{2}{q}} \sum_{x \in \Sigma} \left(\frac{1+(-1)^{h_i(x)}}{2} \right) |x\rangle$$

$$= \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{q}} \sum_{x \in \Sigma} |x\rangle \right) + \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{q}} \sum_{x \in \Sigma} (-1)^{h_i(x)} |x\rangle \right)$$

$$\downarrow$$

$$F|h_i^{-1}(0)\rangle \approx \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} \left(\frac{1}{q} \sum_{xy} \omega^{xy} (-1)^{h_i(x)} |y\rangle \right)$$

most coeffs. here have $|\cdot| \approx \frac{1}{\sqrt{q}}$

So $F|\psi_h\rangle = \sum \underbrace{\hat{\beta}_y |y\rangle}$

"most" y have about $\frac{n}{2}$ 0's and $\frac{n}{2}$ nonzero entries

So we can implement uncomputation if $C^{\perp}$ can handle γ random errors up to weight $\approx \frac{n}{2}$!

In the end, turns out a particular type of Reed-Solomon code satisfies the properties we want.

The YZ Algorithm more abstractly:

The task we're trying to solve is to simultaneously satisfy a linear system (easy!)
+ a non-linear constraint (easy!)

The "entrywise multiplication" trick reduces this to solving a decoding task for the dual of the linear system.

Regev '05: Want to find $x \in \ker(A)$ over $\mathbb{Z}_q^n$ (SIS)
s.t. $\forall i,\ |x_i| \leq$ small

Reduce to decoding: given $y = u + e$, $u \in \ker(A)^\perp$, e a low weight
find e (LWE)

DQI (Jordan et al '24): Want to find $x \in C$ maximizing
$\sum_{i=1}^{n} f_i(x_i)$ ← constraint

9/29/26

Lecture 6: Intro to QMA

We will now transition to talking about quantum proof systems. The question of what you can prove with a quantum state turns out to be very rich, opening up cool connections to physics, cryptography, and beyond.

Recall

NP: Class of languages where there's an efficiently checkable witness. If $L \in NP$, $\exists$ poly-time deterministic TM V s.t.

Completeness: $x \in L \Rightarrow \exists w$ s.t. $V(x,w) = 1$

Soundness: $x \notin L \Rightarrow \forall w \quad V(x,w) = 0$

MA: "Merlin Arthur"

If $L \in NP$, $\exists$ poly-time randomized TM V s.t.

$x \in L \Rightarrow \exists w$ s.t. $\Pr[V(x,w)=1] \geq 2/3$

$x \notin L \Rightarrow \forall w \quad \Pr[V(x,w)=1] \leq 1/3$

QMA: Quantum Merlin Arthur

If $L \in$ QMA, $\exists$ polytime quantum alg. V s.t.

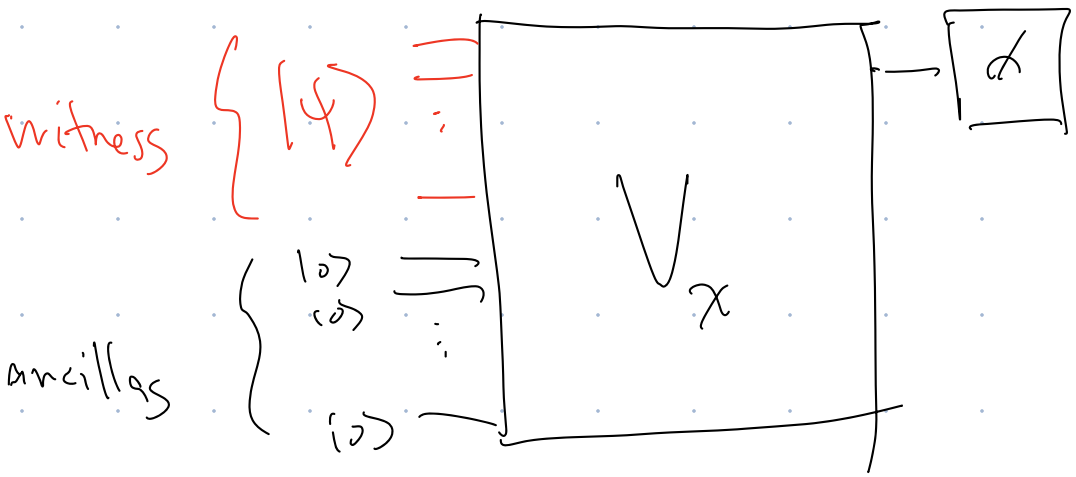
$x \in L \Rightarrow \exists |\psi\rangle$ s.t. $\Pr[V(|x\rangle \otimes |\psi\rangle) = 1] \geq 2/3$

$x \notin L \Rightarrow \forall |\psi\rangle \quad \Pr[V(|x\rangle \otimes |\psi\rangle) = 1] \leq 1/3$

It's obvious that BQP $\subseteq$ QMA and

NP $\subseteq$ MA $\subseteq$ QMA.

QMA is closely connected to eigenvalues. This perspective will let us understand some complexity upper bounds as well as interesting problems in QMA.



$$\Pr[V_x(|\psi\rangle)=1] = \langle\psi|\langle 0^q| V_x^\dagger \left(|1\rangle\langle 1| \otimes I\right) V_x |\psi\rangle |0^q\rangle$$

$$= \langle\psi| M_x |\psi\rangle$$

$$0 \le M \le I$$

$\{M, I-M\}$ form a two-outcome POVM.

This implies that $\max_{|\psi\rangle} \Pr[V(|\psi\rangle)=1]$

$$= \lambda_{max}(M)$$

by the variational characterization of eigenvalues.

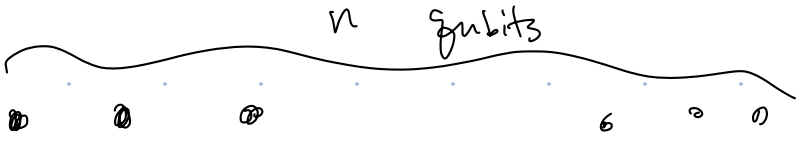
Amplification: Eigenvalues and eigenvectors tensorize, so $\lambda_{max}(M_1 \otimes M_2) = \lambda_{max}(M_1)\lambda_{max}(M_2)$

$\Rightarrow$ Naive amplification by parallel repetition works for QMA $\Rightarrow$ $2/3$, $1/3$ are arbitrary

The Local Hamiltonian Problem:

QMA is all about quantum states that are not necessarily easy to find but whose properties are easily measurable

An example is low-energy states of a quantum many-body system.



$$H = \sum_i h_i$$

$h_i \geq 0$

$\|h_i\| \leq 1$

"local term" acting on k qubits

k-local Hamiltonian

LH problem: Given k-local Hamiltonian H, decide

YES: $\exists\ |\psi\rangle$ s.t. $\langle\psi|H|\psi\rangle \leq a$

NO: $\forall\ |\psi\rangle$ $\langle\psi|H|\psi\rangle \geq b$

Note: this is a promise problem, not a language.

Claim: k-LH problem is in QMA for $k = O(1)$ and $b - a \geq \frac{1}{\text{poly}(n)}$.

Pf: Here's a basic verifier:

- pick a term h_i at random
- measure $|\psi\rangle$ in the eigenbasis of h_i to get outcome μ and accept w/ prob. μ.

Accepts w/ prob. $\langle\psi|h_i|\psi\rangle$

Note: h_i is an $O(1)$-sized matrix so it's efficient to measure in its eigenbasis and compute its eigenvalues.

A fully rigorous analysis would consider the numerical precision / bit complexity of h_i here.

$$\Pr[V(|\psi\rangle)] = \frac{1}{m} \sum_{i \in [m]} \langle\psi|h_i|\psi\rangle = \frac{1}{m}\langle\psi|H|\psi\rangle.$$

This problem is a natural one to study because the ground energy is one of the most basic properties of a physical system. In general in solid-state/condensed matter physics, systems of interest are low-energy.

In-place amplification

We saw how to amplify QMA with parallel repetition. This requires many copies of the witness $|\psi\rangle$. Is there a way to amplify with just one copy?

Marriot & Watrous showed there is! The technique is a powerful idea that shows up in many places in quantum info. (especially in cryptography). The basic ingredient is the following fact about projectors.

Jordan's lemma: Suppose Π_1, Π_2 are projectors. Then we can find a basis where Π_1 & Π_2 are both block-diagonalized into 1×1 & 2×2 blocks.

$$\Pi_1 = \begin{pmatrix} 1 & & & & \\ & 0 & & & \\ & & \alpha & \beta & \\ & & \gamma & \delta & \\ & & & & \ddots \end{pmatrix}, \quad \Pi_2 = \begin{pmatrix} 0 & & & & \\ & 0 & & & \\ & & \alpha' & \beta' & \\ & & \gamma' & \delta' & \\ & & & & \ddots \end{pmatrix}$$

Pf: The idea is to induct. We will "pull out" one block at a time.

Claim: Let $|\alpha\rangle$ be an eigenvec of $\Pi_1\Pi_2$. Then $V = \text{span}\{|\alpha\rangle, \Pi_2|\alpha\rangle\}$ is an invariant subspace for Π_1, Π_2.

(This means for any $|v\rangle \in V$, $\Pi_1|v\rangle$ and $\Pi_2|v\rangle$ are also contained in V)

The claim implies also that $V^\perp$ is an invariant subspace. So we can split the space as $V \oplus V^\perp$ and induct.

Now to prove the claim:

Suffices to show that $\Pi_1|\alpha\rangle, \Pi_1\Pi_2|\alpha\rangle \in V$.

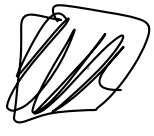
Let $(\Pi_1 + \Pi_2)|\alpha\rangle = \lambda|\alpha\rangle$.

Then $\Pi_1|\alpha\rangle = \lambda|\alpha\rangle - \Pi_2|\alpha\rangle$

$\Rightarrow \Pi_1|\alpha\rangle \in V$

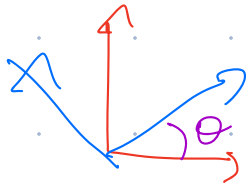
$\Pi_1\Pi_2|\alpha\rangle = \Pi_1(\lambda|\alpha\rangle - \Pi_1|\alpha\rangle)$

$= (\lambda - 1)\Pi_1|\alpha\rangle \in V$



A useful way to think of the blocks is in terms of angles. The 1×1 blocks are where Π_1 and Π_2 are "aligned". On the 2×2 blocks, can further rotate to diagonalize Π_1

$$\Pi_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \Pi_2 = \begin{pmatrix} \cos^2\theta & \cos\theta \sin\theta \\ \cos\theta \sin\theta & \sin^2\theta \end{pmatrix}$$

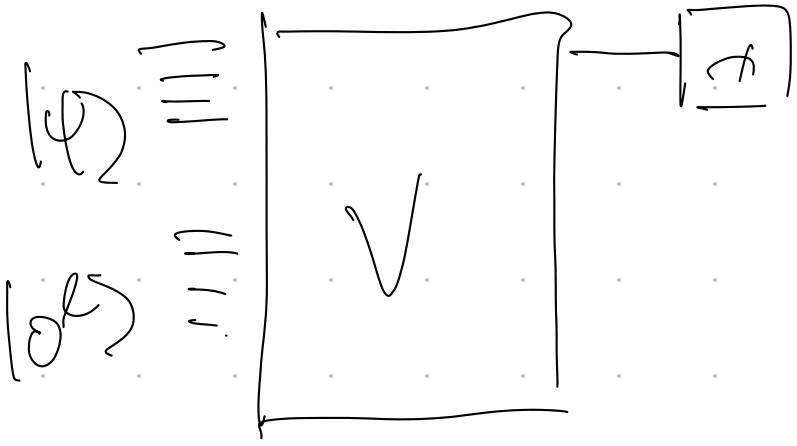


This means that we can understand the eigenvalues of $\Pi_2 \Pi_1 \Pi_2$, say, in terms of the angles.

$$\lambda_{max}(\Pi_2 \Pi_1 \Pi_2) = \max_i \cos^2 \theta_i$$

(i: ith block; $\max$ over blocks)

Armed with this fact we present the MW amplification scheme:



$$\Pi_1 = V^\dagger (|p\rangle\langle 1| \otimes I) V$$

acceptance projector

$$\Pi_2 = I \otimes |0^\ell\rangle\langle 0^\ell|$$

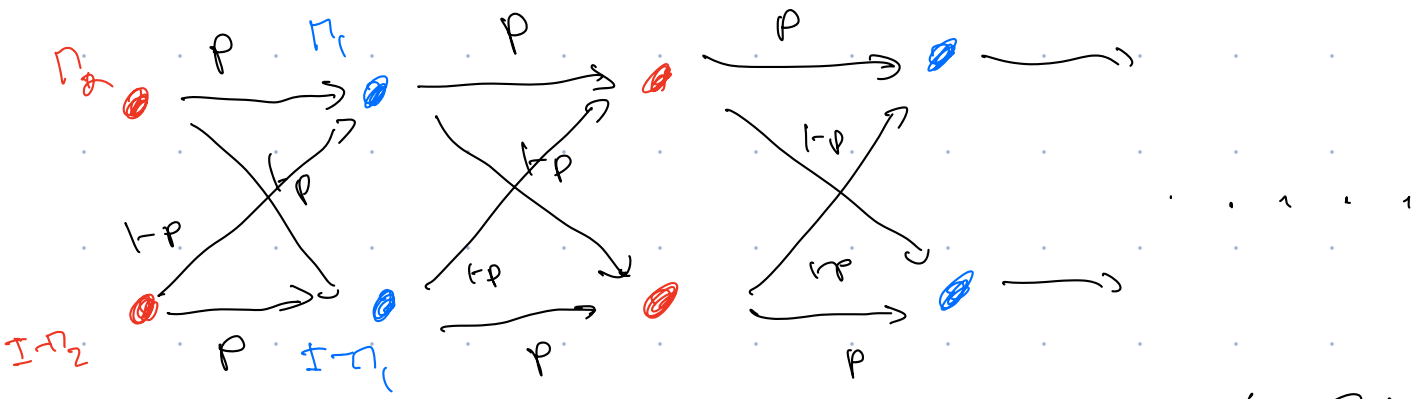
reset ancillas

$$\max_{|\psi\rangle} \Pr[V(|\psi\rangle) = 1] = \lambda_{max}(\Pi_2 \Pi_1 \Pi_2)$$

$$= \cos^2 \theta_{max}$$

The MW algorithm:

Given $|\psi\rangle$, alternately measure in the $\{\Pi_1, I-\Pi_1\}$ basis and the $\{\Pi_2, I-\Pi_2\}$ basis



Suppose $|\psi\rangle$ lies in block i with angle θ_i

$$p := \cos^2\theta_i$$

Counting the number of times the outcome flips from the top row to the bottom row gives us a good estimate of p!

For a general $|\psi\rangle$: observe that Π_1, Π_2 are block diagonal in the Jordan basis, so collapsing $|\psi\rangle$ to a Jordan block commutes w/ the MW procedure!

So mentally can imagine that $|\psi\rangle$ was first measured and collapsed to I block.

10/1/26

Lecture 7: QMA-completeness of Local Hamiltonian

First, a loose end from last time. We claim in-place amplification $\Rightarrow$ QMA $\subseteq$ PP.

Pf: Let M_x be the verifier's measurement on $\mathbb{C}^{2^n}$

Amplify to get $\tilde{M}_x$ on $\mathbb{C}^{2^n}$

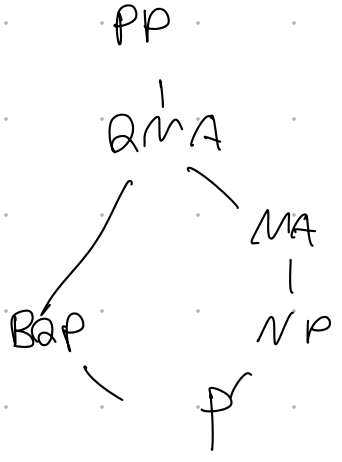
$x \in YES \Rightarrow \lambda_{max}(\tilde{M}_x) \geq 1 - 2^{-2n} \Rightarrow tr(\tilde{M}_x) \geq 1 - 2^{-2n}$

$x \in NO \Rightarrow \lambda_{max}(\tilde{M}_x) \leq 2^{-2n} \Rightarrow tr(\tilde{M}_x) \leq 2^n \cdot 2^{-2n} = 2^{-n}$

$$\Pr\left[\tilde{M}_x \text{ accept on maximally mixed state}\right] = \frac{1}{2^n} tr(\tilde{M}_x) \geq 2^{-n}(1 - 2^{-2n}) \text{ or } \leq 2^{-2n}$$

simulate this using the BQP $\subseteq$ PP path integral technique

So we have the complexity diagram

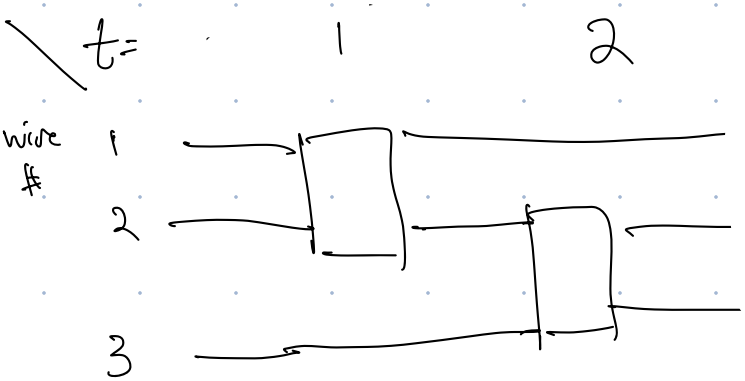


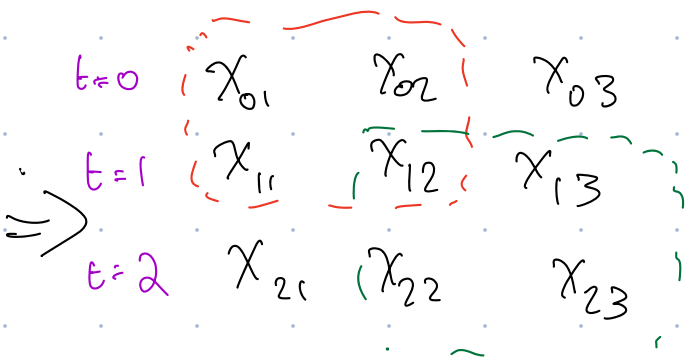
Last time we saw that the local Hamiltonian problem is contained in QMA. Today we will show that it is QMA-hard, and thus QMA-complete.

This will be the quantum analog of the Cook-Levin thm showing that SAT is NP-complete.

Cook-Levin

The idea is to convert the circuit of an NP verifier into a set of local constraints on the "tableau" of that circuit





x_{ti} = bit on wire i at time t

Constraints force x_{ti} to be correct

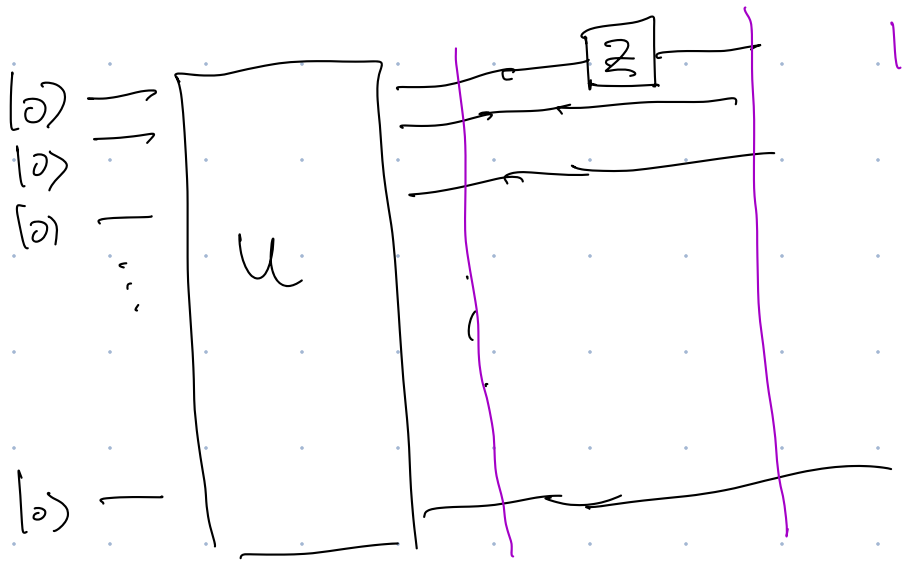
$\exists$ an assignment to the $\{x_{i,j}\}$'s s.t. all constraints are satisfied and input and output bits are correct

$\Uparrow$

$\exists$ a witness making the circuit accept.

Unfortunately, the Cook-Levin tableau idea breaks for quantum circuits, because of local indistinguishability

Imagine a circuit that looks like this



$|\psi_{t+1}\rangle = |CAT^-\rangle$

$= \frac{1}{\sqrt{2}}(|0^n\rangle - |1^n\rangle)$

$|\psi_t\rangle = |CAT^+\rangle = \frac{1}{\sqrt{2}}(|0^n\rangle + |1^n\rangle)$

No local constraints can distinguish $|\psi_t\rangle$ and $|\psi_{t+1}\rangle$, so no way to check that the gate was applied!

Feynman–Kitaev Construction

Instead of a tableau, encode the history of a computation using superposition

$$|\psi_{history}\rangle = \frac{1}{\sqrt{T+1}} \sum_{t=0}^{T} |\psi_t\rangle$$

This solves the local indistinguishability problem.

The state

$$\frac{|CAT_t\rangle|t\rangle + |CAT_{t+1}\rangle|t+1\rangle}{\sqrt{2}}$$

is invariant under $Z \otimes (|t\rangle\langle t+1| + |t+1\rangle\langle t|)$

More generally here's the idea.

$$H = H_{in} + H_{prop} + H_{out}$$

- H_{in}: at $t=0$ ancillas are set to $|0\rangle$
- H_{prop}: propagates computation from $t=0 \to T$
- H_{out}: at $t=T$, output bit is 1.

Each piece will be a sum of PSD local terms $h_i \geq 0$.

Interpret $h_i|\psi\rangle = 0 \Rightarrow$ "ith constraint is satisfied"

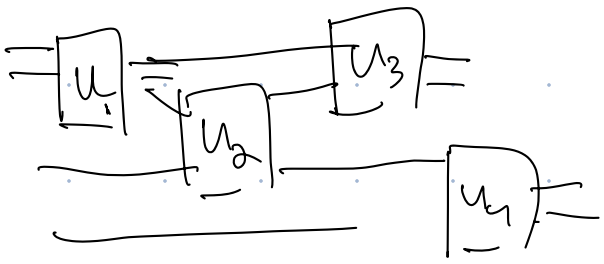
$$H_{in} = \sum_{i \in \text{ancilla}} |1\rangle\langle 1|_i \otimes |0\rangle\langle 0|_{clock}$$

annihilates state if ancilla is correctly set to $|0\rangle$.

$$H_{out} = |0\rangle\langle 0|_{\text{output qubit}} \otimes |T\rangle\langle T|_{clock}$$

annihilates state if output qubit is $|1\rangle$ (= "accept")

The propagation terms are the interesting ones.



U_t = t'th gate in circuit

$$H_{prop} = \sum_t \frac{1}{2}\Big(\underbrace{I \otimes |t\rangle\langle t| + I \otimes |t+1\rangle\langle t+1| - U_t \otimes |t+1\rangle\langle t| - U_t^\dagger \otimes |t\rangle\langle t+1|}_{H_{prop,t}} \Big)$$

For intuition, write $H_{prop,t}$ restricted to the subspace where the clock is $|t\rangle$ or $|t+1\rangle$ (it's 0 outside this subspace anyway).

$$H_{prop,t} = \frac{1}{2} \begin{array}{c} \\ t \\ t+1 \end{array} \begin{array}{c} \begin{array}{cc} t & t+1 \end{array} \\ \begin{pmatrix} I & -U_t^\dagger \\ -U_t & I \end{pmatrix} \end{array}$$

$$= \frac{1}{2} R_t^\dagger \begin{pmatrix} I & -I \\ -I & I \end{pmatrix} R_t, \qquad R_t = \begin{pmatrix} I & 0 \\ 0 & U_t \end{pmatrix}$$

$$= R_t^\dagger \left(I \otimes \text{"}|-\rangle\langle -|\text{"}_{clock} \right) R_t$$

$$\text{"}|\pm\rangle\text{"}_{clock} = \frac{|t\rangle \pm |t+1\rangle}{\sqrt{2}}$$

This means that $H_{prop,t}$ is a projector: things that satisfy it are in the kernel of the projector. We can read this out

$$\ker(H_{prop,t})\Big|_{t,t+1} = \text{span}\left\{ R_t \left(|\psi\rangle \otimes \text{"}|+\rangle\text{"}_{clock} \right) : \text{any } |\psi\rangle \right\}$$

restrict to $t, t+1$ subspace on clock

$$= \text{span}\left\{ \frac{1}{\sqrt{2}} \left(|\psi\rangle|t\rangle + U_t|\psi\rangle|t+1\rangle \right) \right\}$$

propagation of t'th gate!

Putting it together, you get

$$\ker(H_{prop}) = \text{span}\left\{ \frac{1}{\sqrt{T+1}} \sum_{t=0}^{T} U_t \dots U_1 |\psi\rangle \otimes |t\rangle \right\}$$

any state $|\psi\rangle$

We are now ready to sketch the proof of QMA-hardness

Thm: Local Hamiltonian is QMA-hard

Pf: Given QMA verifier V, input x, let V_x be the circuit with x hardcoded.

amplified

$$H_{V,x} = H_{in} + H_{prop} + H_{out}$$

Claim: if $x \in YES$, $\lambda_{min}(H_{V,x}) \leq a = \exp(-n)$

if $x \in NO$, $\lambda_{min}(H_{V,x}) \geq b = \Omega\left(\frac{1}{T^2}\right)$

Pf of claim: For the completeness/YES case, to show $\lambda_{min}(H) \leq a$, it suffices to exhibit $|\varphi\rangle$ s.t. $\langle\varphi|H|\varphi\rangle \leq a$. Taking

$$|\varphi\rangle = \frac{1}{\sqrt{T+1}} \sum_{t=0}^{T} U_t \cdots U_1 \left(|\psi\rangle \otimes (0^e) \otimes |t\rangle \right)$$

($|\psi\rangle$: optimal QMA witness)

does the job!

$$\langle\varphi|H|\varphi\rangle = \langle\varphi|H_{in}|\varphi\rangle + \langle\varphi|H_{prop}|\varphi\rangle + \langle\varphi|H_{out}|\varphi\rangle$$

$$= 0 + 0 + \frac{1}{T+1} \Pr\left[V \text{ rejects } |\psi\rangle \right]$$

$$\leq \exp(-n) \text{ by amplification.}$$

The soundness/NO case is a bit harder!

WTS $\forall |\psi\rangle$, $\langle\psi|H|\psi\rangle \geq \Omega\left(\frac{1}{T^2}\right)$.

$$H = (H_{in} + H_{prop}) + H_{out}$$

minimum nonzero eigenval is $\Omega(\frac{1}{T^2}) =: \lambda_1$

minimum nonzero eigenval is 1.

If $|\psi\rangle$ has energy much less than $\frac{1}{T^2}$, we are done!

IF $|\psi\rangle$ has energy much smaller than λ_1, then it has overlap with the ground space of $H_{in} + H_{prop}$ $\Rightarrow$ has high energy on H_{out} because all computational histories lead to rejection.

What about locality?

The Hamiltonian we wrote is unfortunately not local, at least if we encode the clock in binary.

$|t+1\rangle\langle t|$

naively, acts on $\log T$ qubits.

The solution is to encode the clock in unary:

$$|t\rangle = |\underbrace{11\ldots1}_{t\ 1's}0\ldots0\rangle$$

For full details see the typed notes from last year + attached references

This tells us that 5-local Hamiltonians on qubits are QMA-complete. This has been extended to many different families: e.g. 2-local H on qubits is also QMA-complete.

A different perspective on Hamiltonians

Recall the Schrödinger equation

$$i\hbar \frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle$$

$$|\psi(t)\rangle = \underbrace{e^{-iHt/\hbar}}_{\text{unitary}} |\psi(0)\rangle$$

A Hamiltonian is not just a static object (a system of constraints) but also a dynamic one (describing time evolution)

Feynman came up with (essentially) the Feynman-Kitaev construction to show how a Hamiltonian evolution could simulate a computation.

$$2H_{prop} = \begin{pmatrix} \ddots & & & \\ & -1\ 2\ -1 & & \\ & & -1\ 2\ -1 & \\ & & & -1\ 2\ -1 \\ & & & & \ddots \end{pmatrix}$$

take the trivial circuit where all gates are identity

It turns out this is essentially a discrete version of the Laplacian operator

$$|\psi\rangle = \sum_x \psi(x) |x\rangle$$

$$\partial \psi(x) := \psi(x+1) - \psi(x) \quad \leftarrow \text{discrete derivative}$$

$$\partial^T \psi(x) := \psi(x-1) - \psi(x)$$

$$\partial^T \partial \psi = \partial \left(\psi(x+1) - \psi(x) \right)$$

$$= \left(\psi(x) - \psi(x-1) \right) - \left(\psi(x+1) - \psi(x) \right)$$

$$= 2\psi(x) - \psi(x-1) - \psi(x+1)$$

$$= H_{prop}\, \psi$$

So we can understand H_{prop} intuitively by looking at the "continuum limit"

$$H_{prop} \to \partial^T \partial = -\frac{d^2}{dx^2}$$

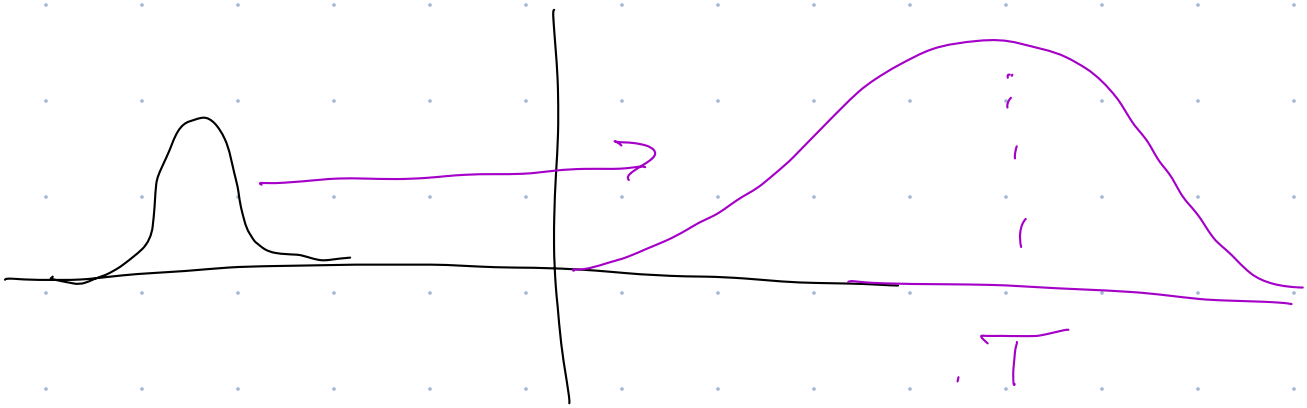
Eigenstates are plane waves $|\hat{k}\rangle := \sum_x e^{ikx} |x\rangle$

$$H_{prop} |\hat{k}\rangle = k^2 |\hat{k}\rangle$$

$$e^{-iH_{prop}t} |\hat{k}\rangle = \sum_x e^{i(kx - k^2 t)} |x\rangle$$

wave moving to the right at speed k.

So to simulate a computation, prepare a wave packet centered around some negative t and evolve it under H.



As the state evolves under H, time advances forward and gates are applied. Finally, measure and if the clock is in state T, you can read out the answer to your computation.

10/6/27

Lecture 8: QMA vs QCMA

Remember the definitions of MA & QMA

MA: $\exists$ classical verifier V s.t.

$x \in L_{YES} \Rightarrow \exists\, w \in \{0,1\}^{poly(n)}$

$$\Pr[V(x,w) = 1] \geq 2/3$$

$x \in L_{NO} \Rightarrow \forall\, w \in \{0,1\}^{poly(n)}$

$$\Pr[V(x,w) = 1] \leq 1/3$$

QMA: $\exists$ quantum verifier V s.t.

$x \in L_{YES} \Rightarrow \exists\, |\psi\rangle$

$$\Pr[V(x,|\psi\rangle) = 1] \geq 2/3$$

$x \in L_{NO} \Rightarrow \forall\, |\psi\rangle$

$$\Pr[V(x,|\psi\rangle) = 1] \leq 1/3$$

In defining QMA, we made both the verification procedure and the witness quantum.

Could have done something in-between:

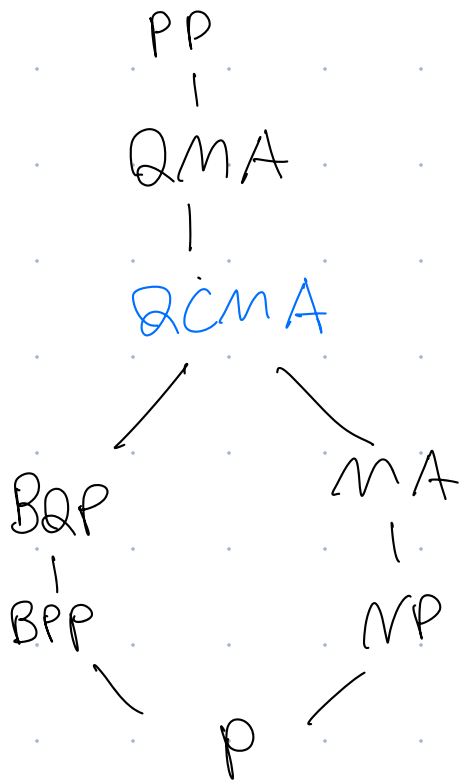
QCMA: $\exists$ quantum verifier V s.t.

$$x \in L_{YES} \Rightarrow \exists\, w \in \{0,1\}^n \quad \Pr[V(x,w)=1] \geq 2/3$$

$$x \in L_{NO} \Rightarrow \forall\, w \in \{0,1\}^n \quad \Pr[V(x,w)=1] \leq 1/3$$

Why do this? Well we know that quantum algorithms can do more than classical algos even on classical input (e.g. factoring). Is the power of QMA coming from this, or from the information in the q. witness itself?

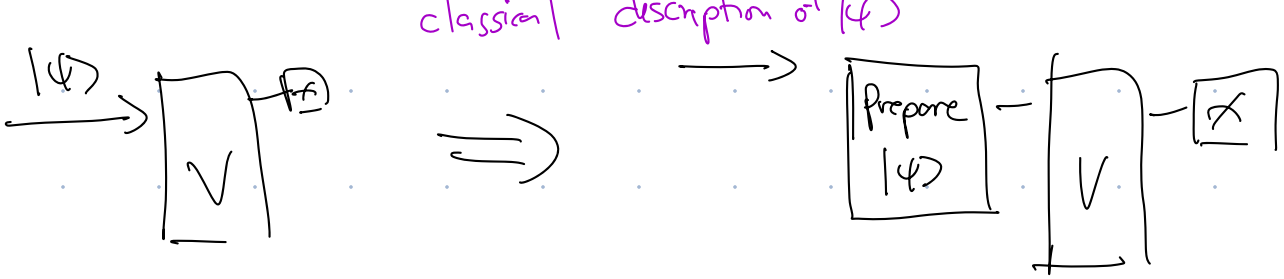
More generally, we want to understand what differentiates NP from QMA.



Conjecture: $QMA \neq QCMA \neq NP$

The intuition behind this conjecture comes from thinking about classical descriptions of quantum states.

The only natural strategy for showing $QMA = QCMA$ would be to show that any QMA protocol can be simulated in QCMA



Ground states of local Hamiltonians seem to be hard to describe classically

- Can be highly entangled
- Don't seem to have poly-size preparation circuits

But proving this is difficult!

What if we ask for a separation in the query/oracle model? This doesn't sound so different from separations we've already shown (BQP vs. NP), but it was actually an open problem for nearly 20 years!

Aaronson - Kuperberg 2006

Idea: $NP^{\mathcal{O}} \not\subseteq BQP^{\mathcal{O}}$ ∈ "a quantum alg can't find a hidden string in a search space of size 2^n, but in NP, the prover can give it to you"

So to show $QMA^O \not\subseteq QCMA^O$, hide a "hidden state" in O that a classical message won't help you find!

To implement this AK considered a more general type of ~~oracle~~: a unitary/quantum oracle

"Classical oracle" $O|x\rangle|b\rangle = |x\rangle|b \oplus f(x)\rangle$

"Quantum oracle" $U|\psi\rangle = |\psi'\rangle$

arbitrary unitary

AK's hard family: $U_\psi = I - 2|\psi\rangle\langle\psi|$ YES

vs.

$U_0 = I$ NO

This decision task is easily solvable by a QMA protocol: just ask for $|\psi\rangle$ and measure U_ψ on it

How about showing a QCMA lower bound?

To start: a BQP lower bound follows from BBBV-type arguments:

$|\Phi^t\rangle :=$ state after t queries to the I oracle

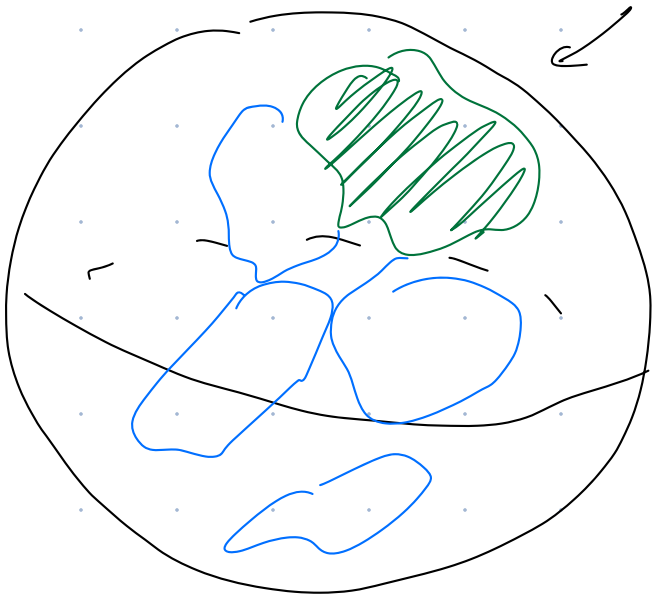
$$\mathop{\mathbb{E}}_{\psi \sim \mu} \| U_\psi |\Phi^t\rangle - |\Phi^t\rangle \| \leq 2 \sqrt{\mathop{\mathbb{E}}_{\psi \sim \mu} \langle \psi | \rho_t | \psi \rangle}$$

"progress" in $t+1^{st}$ query

state of query register in $|\Phi^t\rangle$

This is $\frac{1}{2^n}$ for any possible ρ_t

In the QCMA case, the witness makes things more complicated!



unit sphere over $\mathbb{C}^{2^n}$ (schematic)

Each possible witness $w \in \{0,1\}^m$ is optimal for some subset of states $|\psi\rangle$ (possibly none)

Let w^* be the most popular witness, and S_{w^*} be the states it's optimal for.

$$\Pr_{\psi_n \text{ uniform}}\left[\,|\psi\rangle \in S_{w^*}\,\right] \geq \frac{1}{2^m}$$

↑ by pigeonhole

Key Lemma [AK]:

For any fixed state ρ

$$\mathop{\mathbb{E}}_{\psi_i \in S_{w^*}} \langle\psi_i|\rho|\psi_i\rangle \leq \frac{m+1}{2^n}$$

$\Rightarrow$ On w^*, progress per step is $2\sqrt{\frac{m+1}{2^n}}$ $\Rightarrow$ need $\Omega\left(\sqrt{\frac{2^n}{m+1}}\right)$ queries

This shows that this problem is not in QCMA^U.

Why is this not satisfactory? AK is essentially a counting argument. It says you can't map all the $\exp(2^n)$ different states to 2^m bit strings if $m = o(2^n)$.

But in reality, even optimal witness state $|\psi\rangle_x$ for an instance x of a QMA problem can be mapped to a poly-length string — namely x itself!

So this oracle separation is missing the point!

Several subsequent works tried to improve this result to classical oracles:

FK'16, NN'24, LLPY'23...

But the problem was only solved in 2025!

+ simpler solution in 2026

Bostanci Haferkamp Nirkhe Zhandry '25

Builds on an observation of Zhandry '24:

A QCMA witness can be cloned

but a QMA witness may not be clonable!

So if a QCMA verifier can solve some search problem, it can be rerun many times on the same witness to find many valid outputs.

Example: Set Size Estimation

Suppose the oracle contains a set of marked strings $S \subseteq \{0,1\}^n$. We want to decide if T is small or large.

YES: $|S| \geq 2^{n/2}$

NO: $|S| \leq 2^{n/3}$

We know that a BQP alg. cannot solve unstructured search $\Rightarrow$ to even find one point in S, need a hint from the witness!

Claim: Suppose we have a t-query QCMA alg that solves set size estimation.

For S a YES instance, let w_S be its optimal witness.

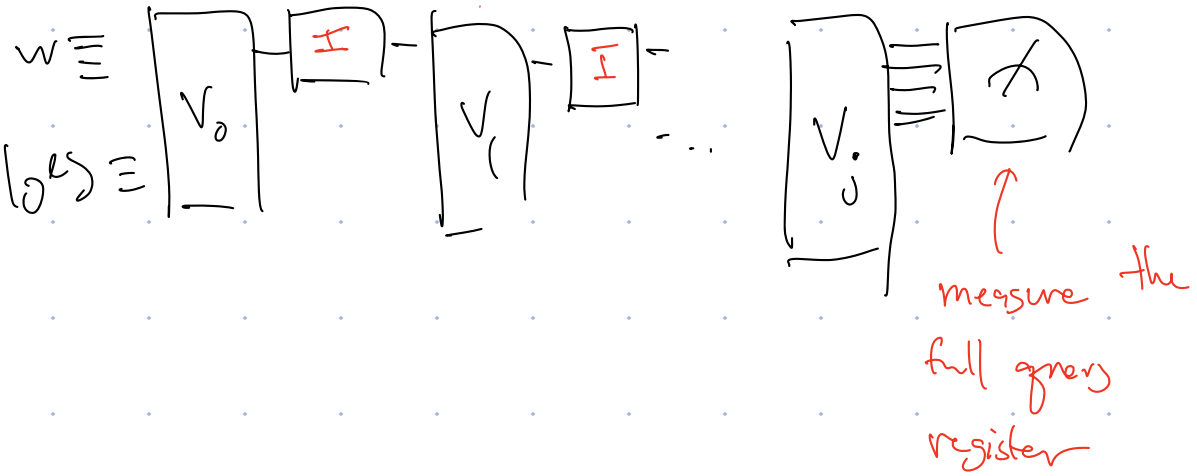
Then $\exists$ a 0-query alg BasicSampler s.t.

$\forall S \in$ YES, $\Pr[\text{BasicSampler}(w_S) \in S] \geq \Omega\left(\frac{1}{t^2}\right)$

Pf: Let the QCMA alg be

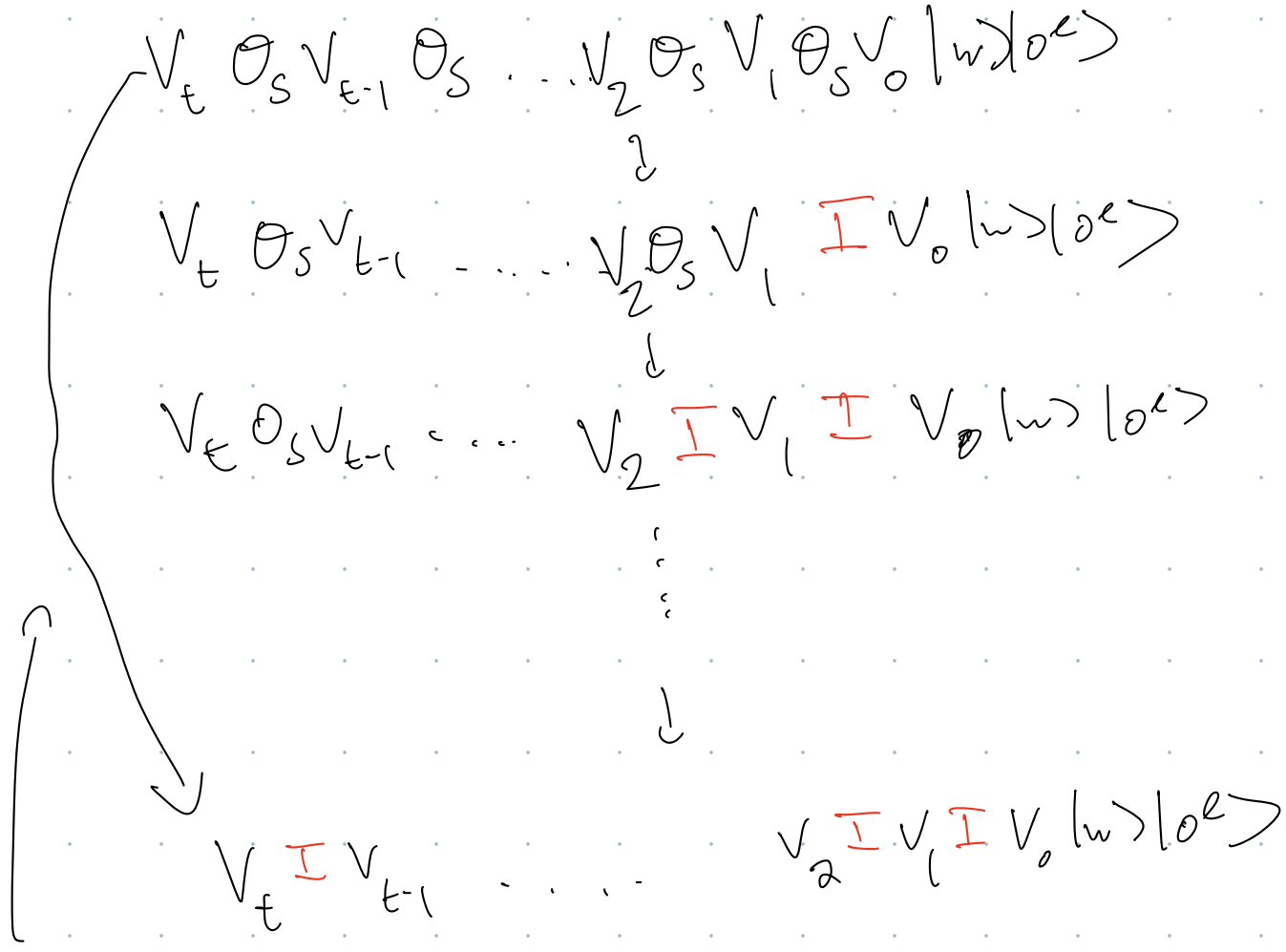
Then BasicSample(w) does the following

- sample $j \in \{0, \ldots, t-1\}$ uniformly.
- Run



We know that our QCMA alg distinguished O_S from $O_\phi = I$.

Therefore,



$\Omega(1)$-far in Euclidean distance

$$\Rightarrow \mathop{\mathbb{E}}_{j \in \{0..t-1\}} \left\| (\Theta_S - I) \underbrace{V_j \otimes I \cdots V_0 |w\rangle|0^\ell\rangle}_{|\psi_j\rangle} \right\| \geq \Omega\left(\frac{1}{t}\right)$$

$$\Rightarrow \mathop{\mathbb{E}}_{j \in \{0..t-1\}} \Pr\left[\text{Measure } |\psi_j\rangle \to x \in S\right] \geq \Omega\left(\frac{1}{t^2}\right)$$

(by Jensen's ineq.)

Can modify this alg to get v distinct samples!

Cumulative Sampler (w, v):

Set $\Delta \leftarrow \emptyset$.

For $i = 1 \dots v$: Δ ← Basic Sampler with Θ_Δ instead of I

Run Basic Sampler (w) to get x_i

If $x_i \in \Delta$, abort.

Else, $\Delta \leftarrow \Delta \cup \{x_i\}$

Output Δ

$$\Pr\left[\text{Cumulative Sampler}(w_S, v) \text{ outputs } v \text{ distinct } x_1 \dots x_v \in S\right]$$

$$= \Omega\left(\left(\frac{1}{t^2}\right)^v\right)$$

$$\Rightarrow \mathop{\mathbb{E}}_{w \in \{0,1\}^m} \Pr[\text{“}\text{”}] = \Omega\left(\frac{1}{2^m}\left(\frac{1}{t^2}\right)^v\right)$$

(This assumes $v \leq 2^{n/3} \Rightarrow \Delta$ is always a NO instance)

Note: this algo makes no queries to S at all!

So it has no business being very good:

$$\Pr\left[\begin{array}{c}\text{any 0-query alg finds}\\ v \text{ distinct points in } S\end{array}\right] \leq O\left(\left(\frac{|S|}{2^n}\right)^v\right)$$

We conclude that

$$\frac{1}{2^m}\left(\frac{1}{t^2}\right)^v \leq K \cdot \left(\frac{1}{2^{n/2}}\right)^v \quad \text{for } v \leq 2^{n/3}$$

(K: some constant)

$$-m - 2v\log t \leq \log K - \frac{vn}{2}$$

$$\Rightarrow \quad m \geq v\left(\frac{n}{2} - 2\log t\right) - \log K$$

($v \geq 2^{n/3}$)

If $m = \text{poly}(n)$, then t must be superpoly!

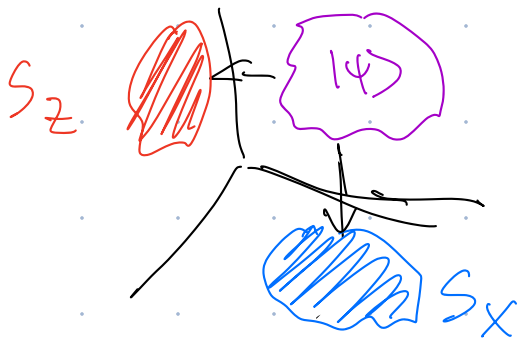
(on the order of $2^{n/4}$)

So we have shown that Set Size Estimation is not in QCMA!

Unfortunately, it's not in QMA either [Kretschmer]

BHNZ'25: Create a problem with a similar flavor that's in QMA by design.

"Spectral Correlation":



For a state $|\psi\rangle$, think of S_X and S_Z as the set of "heavy points" of $|\psi\rangle$ measured in X basis and Z basis

"X-shadow" and "Z-shadow" of state.

Problem: Given oracles for S_X, S_Z, decide if

YES: $\exists\ |\psi\rangle$ s.t. shadows approximately match S_X, S_Z

NO: no such $|\psi\rangle$ exists.

Given a state $|\psi\rangle$, it's easy to construct S_X, S_Z that are a YES instance.

Consider $|\psi\rangle$ s.t. S_Z is pretty small.

Then can make a No instance

$|\psi\rangle$

S_X S_Z

S'_X = tiny subset of S_X, S_Z

Distinguishing (S_X, S_Z) from (S'_X, S_Z) has the flavor of Set Size Estimation. But now it can be done in QMA: just ask for $|\psi\rangle$ and measure it in X-basis or Z-basis.

To show QCMA lower bound, show that a QCMA alg

$\Rightarrow$ an alg for finding

r distinct points in S_X

Technique used by BHNZ is a sophisticated variant of compressed oracle!

Spectral Forrelation & Hamiltonians

Intuitively, SF is an oracle version of "XZ Hamiltonians", a QMA-complete family of Hamiltonians

$$H = H_X + H_Z$$

H_X: diagonal in X basis

H_Z: diagonal in Z basis

Bostanci Huang Vaikuntanathan '26

took this class last year!

Came up with an alternate separation based on YZ problem!

"Code Intersection Subset Size:"

For $E \subseteq \{0,1\}^n \times \Sigma^n$, given oracle

$$O[H, E](x, v) = \begin{cases} 1 & \text{if } v \in C,\ H(x) = x,\ (x, v) \in E \\ 0 & \text{otherwise} \end{cases}$$

$H = (h_1, h_2, \ldots, h_n)$

Decide if

YES: $|E| = 2^n$

NO: $|E| \ll 2^n$

Intuitively, use code intersection to encrypt access to E.

Note that unlike YZ, no longer get access to H directly!

Yet problem is still in QMA:

ask for witness state

$$\left(|h_1^{-1}(0)\rangle \otimes |h_1^{-1}(1)\rangle\right) \otimes \left(|h_2^{-1}(0)\rangle \otimes |h_2^{-1}(1)\rangle\right) \otimes \ldots$$

QCMA lower bound is by a version of the Set Size Estimation argument above!

10/8/26

Lecture 9: QMA(2) and the power of unentanglement

Last time we saw that the QMA-QCMA separation suggests that the power of QMA comes from superposition & entanglement in the witness state.

Today we will see a "variant" of QMA whose power comes from unentanglement

QMA(2): ∃ quantum polytime verifier V taking as input a witness

$$|\psi\rangle_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$$

s.t.

$x \in YES \Rightarrow \exists \; |\psi\rangle_{AB} = |\alpha\rangle_A \otimes |\beta\rangle_B$

s.t. $\Pr[V(x, |\psi\rangle_{AB}) = 1] \geq 2/3$

$x \in NO \Rightarrow \forall \; |\psi\rangle_{AB} = |\alpha\rangle_A \otimes |\beta\rangle_B$

$\Pr[V(x, |\psi\rangle_{AB}) = 1] \leq 1/3$

Note that both completeness and soundness hold only for product states.

(Can imagine there are "two quantum Merlins")

We have the immediate containment

$QMA \subseteq QMA(2)$

(just ignore the B system)

But upper bounds like $QMA(2) \overset{?}{\subseteq} PP$ or EXP are not obvious! In fact best known upper bound is

$$QMA(2) \subseteq NEXP.$$

Fundamentally this is because $QMA(2)$ is about a more challenging optimization problem than QMA

$$QMA: \max_{|\psi\rangle} \Pr[V(|\psi\rangle)] = \lambda_{max}(M)$$

$$QMA(2): \max_{|\psi\rangle} \Pr[V(|\psi\rangle)] = \max_{|\alpha\rangle|\beta\rangle} \langle\alpha|\langle\beta|M|\alpha\rangle|\beta\rangle$$

$$:= h_{sep}(M)$$

These quantities can be far apart!

$M \in B(\mathcal{H}_A \otimes \mathcal{H}_B)$

Say these are both $\mathbb{C}^d$

Ex: Take $M = |\Phi\rangle\langle\Phi|$

$$|\Phi\rangle = \frac{1}{\sqrt{d}} \sum_{i=1}^{d} |i\rangle|i\rangle$$

maximally entangled state

$$h_{sep}(M) = \max_{|\alpha\rangle,|\beta\rangle} \langle\alpha|\langle\beta|\Phi\rangle\langle\Phi|\alpha\rangle|\beta\rangle$$

$$= \max_{|\alpha\rangle|\beta\rangle} |\langle\Phi||\alpha\rangle|\beta\rangle|^2$$

For any $|\alpha\rangle$, let U_α be the unitary mapping $|1\rangle$ to $|\alpha\rangle$. Then

$$|\Phi\rangle = U_\alpha \otimes \overline{U}_\alpha \, |\Phi\rangle$$

(✓ complex conjugate)

$$= \frac{1}{\sqrt{d}} |\alpha\rangle \otimes |\overline{\alpha}\rangle + \text{orthogonal stuff}$$

$$\Rightarrow \max_{|\alpha\rangle|\beta\rangle} |\langle \Phi | \, |\alpha\rangle|\beta\rangle|^2 = \frac{1}{d}.$$

When $d = 2^n$, this means $h_{sep}^{(n)}$ can be vanishingly small even when $\lambda_{max}^{(n)} = 1$!

The fact that h_{sep} seems to be harder to compute than λ_{max} opens the opportunity to use QMA(2) protocols to solve harder computational problems!

Short Proofs for 3-coloring

The "killer app" for QMA(2) was discovered by Blier & Tapp

Thm: 3COL $\in$ QMA(2) with

completeness	1
soundness	$1 - 1/\text{poly}(n)$
witness size	$O(\log n)$ qubits

In contrast: Cannot have a QMA protocol for 3COL w/ $O(\log n)$-size witnesses unless $NP \subseteq BQP$

Claim: $QMA_{\text{log-sized witness}} = BQP$

qubits in witness $\rightarrow m = O(\log n)$

Pf: Amplify in-place w/ Marriott-Watrous!

$$\Pr_{\text{random } |\psi\rangle}[V(|\psi\rangle)] \geq \frac{1}{2^m} \cdot (1 - 2^{-n}) \quad \text{YES}$$

$m = \text{"poly}(n)$

$$\Pr_{\text{random } |\psi\rangle}[V(|\psi\rangle)] \leq 2^{-n} \quad \text{NO}$$

Gap is $\frac{1}{\text{poly}(n)}$ so can be estimated with poly-many repetitions!

Pf of thm

We expect the proof states to look like

$$|\alpha\rangle = |\beta\rangle = \frac{1}{\sqrt{V}} \sum_{v \in V} |v\rangle |c_v\rangle$$

v: vertex label; c_v: color of vertex

"Encode coloring in superposition"

Execute one of the following tests:

1) SWAP test: check that $|\alpha\rangle = |\beta\rangle$

2) Uniqueness test:

measure both states to get

(v_α, c_α) (v_β, c_β)

Check that if $v_\alpha = v_\beta$, then $c_\alpha = c_\beta$

3) Edge test:

measure both states to get

(v_α, c_α) (v_β, c_β)

Check that if $(v_\alpha, v_\beta) \in E$ then

$c_\alpha \neq c_\beta$

4) Uniformity test:

measure the color register of $|\alpha\rangle$ in Fourier basis

$$|\bar{0}\rangle = \frac{1}{\sqrt{3}} \sum_c |c\rangle \quad , \quad |\bar{1}\rangle = \frac{1}{\sqrt{3}} \sum_c \omega^c |c\rangle \quad , \quad |\bar{2}\rangle = \frac{1}{\sqrt{3}} \sum_c \omega^{2c} |c\rangle$$

If get outcome $\bar{0}$, then measure vertex register in Fourier basis

accept if outcome is $|\bar{0}_V\rangle = \frac{1}{\sqrt{V}} \sum_v |v\rangle$

else reject

Else accept

It's easy to check completeness. For soundness, let's do the 0-error case: this has all the main ideas.

0-error soundness: if every test passes w/ prob 1, then $\exists$ a valid 3 coloring of the graph

Pf: SWAP test $\Rightarrow$ $|\alpha\rangle = |\beta\rangle$

Uniqueness test: $|\alpha\rangle = \sum_{u \in V} q_u |u\rangle \left(\sum_c \beta_{u,c} |c\rangle \right)$

$\sum_c |\beta_{u,c}|^2 = 1$

$|\beta\rangle = \ldots - \sum_u |\alpha_u|^2 = 1$

Measure both $\Pr[c_\alpha = c_\beta \mid u_\alpha = u_\beta = u]$

$$= \sum_c |\beta_{u,c}|^4$$

$\Rightarrow$ only one $\beta_{u,c} = 1$

rest are $= 0$

$\Rightarrow$ a unique color for each vertex

Uniformity test: $|\psi\rangle = \sum_u \alpha_u |u\rangle|c_u\rangle$

This is the most fun test!

$$(I \otimes |\bar{0}\rangle\langle\bar{0}|)|\psi\rangle = \sum_u \alpha_u |u\rangle \cdot \left(\frac{1}{\sqrt{3}} |\bar{0}\rangle\right)$$

no phase!

So if you pass the measurement on the vertex register, $\sum_u \alpha_u |u\rangle \propto \frac{1}{\sqrt{V}} \sum_u |u\rangle$

$\Rightarrow$ all α_u are equal!

So every vertex u is equally represented in the state.

Edge test: $|\alpha\rangle = \frac{1}{\sqrt{V}} \sum_u |u\rangle|c_u\rangle$

$|\beta\rangle = \frac{1}{\sqrt{V}} \sum_u |u\rangle|c_u\rangle$

$\Downarrow$ induces a coloring for the vertices!

Passing edge test $\Rightarrow$ coloring is admissible ∎

Note that the unentanglement between the two proofs was essential! There's an easy cheating state if we allow entanglement!

$$|\psi\rangle = \alpha \sum_{\substack{v_1 \neq v_2 \\ c_1 \neq c_2}} |v_1 c_1 v_2 c_2\rangle + \beta \sum_{\substack{v \\ c}} |v_1 c_1 v_1 c_1\rangle$$

- SWAP ✓ (symmetric under permutation)
- Uniqueness ✓ $v_1 = v_2 \Rightarrow c_1 = c_2$

- Edge ✓ $v_1 \neq v_2 \Rightarrow c_1 \neq c_2$

- Uniformity:

$$(I \otimes |\bar{0}\rangle\langle\bar{0}| \otimes I \otimes I)\,|\psi\rangle$$

$$= \alpha \sum_{\substack{v_1 \neq v_2 \\ c_1 \neq c_2}} \frac{1}{\sqrt{3}} |v_1\rangle|\bar{0}\rangle|v_2\rangle|c_2\rangle + \beta \sum_{v_1 c_1} \frac{1}{\sqrt{3}} |v_1\rangle|\bar{0}\rangle|v_1\rangle|c_1\rangle$$

$$= \frac{1}{\sqrt{3}} \sum_{v_1 c_1} |v_1\rangle|\bar{0}\rangle \left(\alpha \sum_{\substack{v_2 \neq v_1 \\ c_2 \neq c_1}} |v_2\rangle|c_2\rangle + \beta |v_1\rangle|c_1\rangle \right)$$

$$= \frac{1}{\sqrt{3}} \sum_{v_1} |v_1\rangle|\bar{0}\rangle \left(\beta |v_1\rangle \sum_{c_1} |c_1\rangle + 2\alpha \sum_{v_2 \neq v_1} |v_2\rangle \sum_{c_1} |c_1\rangle \right)$$

every color is counted twice

If we take $\alpha = \frac{\beta}{2}$,

$$|\psi\rangle = \frac{1}{\sqrt{3}} \sum_{v_1} |v_1\rangle|0\rangle \left(\sum_{v_2,c} |v_2\rangle|c\rangle \right)$$

This can be extended to a constant gap with some tricks, if you let the witness be size $\tilde{O}(\sqrt{n})$ ← still smaller than possible with QMA, assuming Exp. Time Hypothesis

For a long time, no oracle separations between QMA & QMA(2) were known.

This August: $QMA(2)^U \not\subseteq QMA^U$ unitary U

October: $QMA(2)^O \not\subseteq QMA^O$ O classical

Both AI generated

10/15/26

Lecture 10: The Quantum PCP Conjecture

The Cook-Levin thm says that classical computations are locally checkable, and Feynman-Kitaev gives us a version of this for quantum computations as well.

But classically, something much stronger is true: classical computations are robustly locally checkable.

PCP thm

Given a "classical local Hamiltonian" (= local CSP)

$$H = \sum_{i=1}^{m} h_i \quad \leftarrow \text{diagonal projector}$$

it is NP-hard to decide whether

YES: $\lambda_{min}(H) = 0$

or

NO: $\lambda_{min}(H) \geq$ m · a (some constant a)

constant fraction of total energy m

This is called the Probabilistically Checkable Proofs theorem because given an assignment $|x\rangle$ to H, can simply sample I term of H at random and measure on $|x\rangle \Rightarrow$

proof system w/ completeness 1

soundness $1-a$

"Any NP computation has a witness that is spot-checkable in $O(1)$ locations"

"NP-hard CSPs remain NP-hard even up to constant-factor approximations"

This theorem is one of the jewels of classical complexity theory!

The quantum PCP conjecture states the analogous thing for general LH:

(still wide open)

Given LH $H = \sum_{i=1}^{m} h_i$ $\in$ $\|h_i\| \leq I$, $h_i \geq 0$

it is QMA-hard to distinguish

Yes: $\lambda_{min}(H) \leq a$

No: $\lambda_{min}(H) \geq b$

$b - a = m \cdot \Omega(1)$

(recall: in Feynman-Kitaev, we got $b - a = 1/\text{poly}(n)$)

This is interesting for the same reasons as classical PCP, but another really interesting thing is what it says about the complexity of classical descriptions of low-energy states

qPCP + QMA ≠ QCMA

$\Rightarrow$ ∃ families of H where all low-energy states cannot be prepared by a poly(n)-sized quantum circuit

qPCP + QMA ≠ NP

$\Rightarrow$ ∃ families of H where all low energy states are highly entangled.

In fact, can deduce something stronger!

qPCP + QMA ≠ NP

$\Rightarrow$ ∃ families of H where there are no low-energy trivial states (NLTS)

Def. A Hamiltonian H has NLTS if $\exists\, c, m$

$\forall\, |\psi\rangle$ s.t. $\langle\psi|H|\psi\rangle \leq \lambda_{min} + c \cdot m$

$|\psi\rangle$ is not prepared by a constant-depth quantum circuit of 2-qubit gates.

NLTS on its own is a structural prerequisite for qPCP — doesn't say anything about QMA-hardness

Freedman & Hastings '13: "find an NLTS family of Hamiltonians at least (before trying to prove qPCP)"

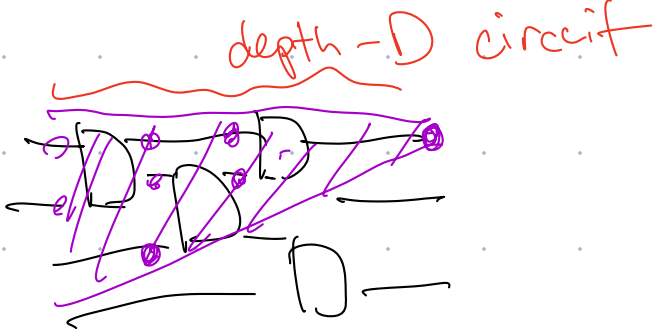
Eldar & Harrow '15: an approach that didn't quite work

Anshu Breuckmann Nirkhe '22: a proof using newly developed quantum codes

NLTS thm is evidence for qPCP, but also a cool structural result about entanglement in locally-checkable states that is interesting in its own right!

Why does qPCP + NP ⊄ QMA ⇒ NLTS?

Lightcone argument



Every qubit in the output has a past lightcone of size at most 2^D

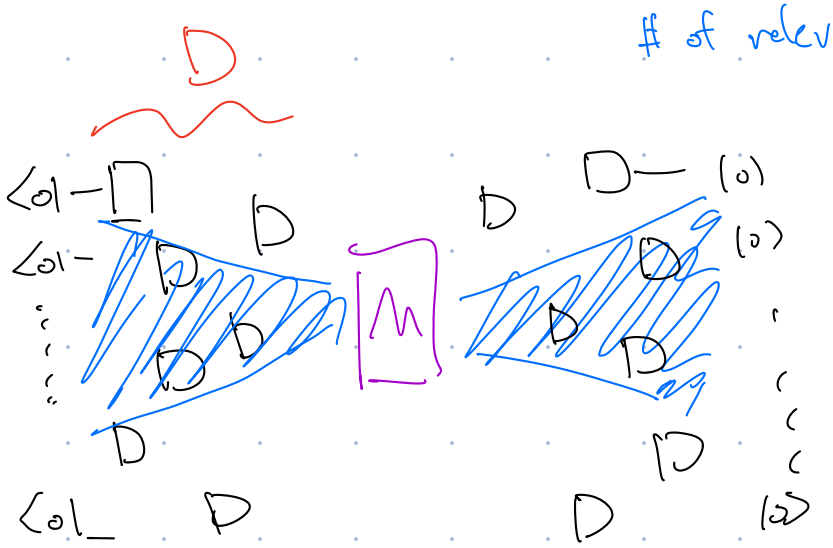
↑ a 2-qubit gate can only expand by a factor of 2.

Similarly, every input qubit has a future lightcone of size at most 2^D.

Suppose $|\psi\rangle$ has a depth-D circuit. Then for every set of k output qubits, can compute the expectation value of any observable M on these qubits in $\exp(k \cdot 2^D)$ time classically!

($k \cdot 2^D$: # of relevant qubits)

$\langle\psi|M|\psi\rangle =$



If $D = O(1)$, this is still constant!

So the NP witness is just the circuit!

Can use it to compute $\langle\psi|H|\psi\rangle$ classically in poly time.

How do we show circuit depth lower bounds?

Fact: The state $|CAT^+\rangle = \frac{1}{\sqrt{2}}\left(|0^n\rangle + |1^n\rangle\right)$

requires depth $\Omega(\log n)$ to prepare.

Pf: Essentially by local indistinguishability.

Suppose U of depth $D < \log n$ prepares $|CAT^+\rangle$

$$|CAT^+\rangle = U|0^n\rangle$$

$$|0^n\rangle = U^\dagger |CAT^+\rangle$$

This implies that $U^\dagger|CAT^-\rangle$ is orthogonal to $|0^n\rangle$.

Define the local Hamiltonian

$$H = \sum_i \underbrace{U \overbrace{(|1\rangle\langle 1|)_i}^{h_i} U^\dagger}_{\text{locality} \cdot 2^D < n}$$

The above implies

$$0 = \langle CAT^+ | H | CAT^+ \rangle \neq \langle CAT^- | H | CAT^- \rangle$$

$$\Rightarrow \exists\ i \quad \text{s.t.} \quad \langle CAT^+ | h_i | CAT^+ \rangle \neq \langle CAT^- | h_i | CAT^- \rangle$$

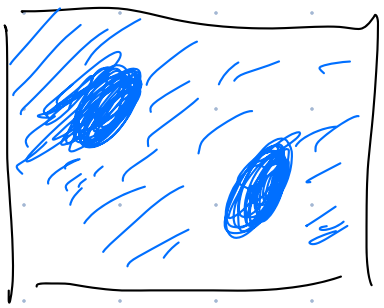
But this is impossible because of local indistinguishability!

This argument unfortunately also tells us that $|CAT^+\rangle$ can never be the unique ground state of a local Hamiltonian, so it can't directly give rise to an NLTS Ham.

But can extract a useful idea:

Lemma [Eldar-Harrow]. If $|\psi\rangle$ measured in the Z basis (or X basis) yields a distribution concentrated on two widely separated clusters of bitstrings, then $|\psi\rangle$ has circuit depth $\Omega(\log n)$.

"fuzzy CAT state"



Pf idea: Use local indistinguishability of relative phases between clusters like in CAT state.

We now want to find a local H s.t. all low-energy states are clustered like this.

First candidate:

H = parent Hamiltonian of a CSS code with low-weight checks

Recall: a CSS code is a quantum code built from 2 classical codes, one in each of X and Z bases.

Logical Z states: for $x \in C_2$

$$\frac{1}{\sqrt{|C_X^\perp|}} \sum_{y \in C_X^\perp} |x + y\rangle$$

↑ cluster centered around $x \in C_2$

Change to X basis ↓

Superposition of clusters around $z \in C_X$

Can show that any CSS code state has the clustering prpety in X or Z basis.

Moreover, CSS code states are 0-error states of a "parent Hamiltonian"

$$H = H_X + H_Z$$

H_X: checks of C_X

H_Z: checks of C_Z

$\Rightarrow$ 0-energy states of H have circuit depth $\Omega(\log n)$

To get NLTS, need to boost this to all low-energy states

Eldar-Harrow: true if the code is "quantum locally testable" (first such code constructed this year)

ABN: true if code has some kind of "poor man's local testability" (obtained from the "good qLDPC codes" of Panteleev & Kalachev)

Lecture 11 notes are available separately online (as "last year's notes")

Lecture 12 is on its way...

10/27/26

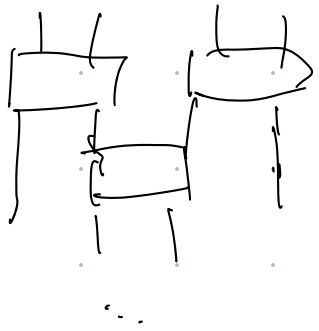
Lecture 13: Fault tolerance & quantum PCP

Last time we saw an approach to qPCP based on "gap amplification": transforming a Hamiltonian to amplify the energy. This is roughly how Dinur '07's proof of PCP goes.

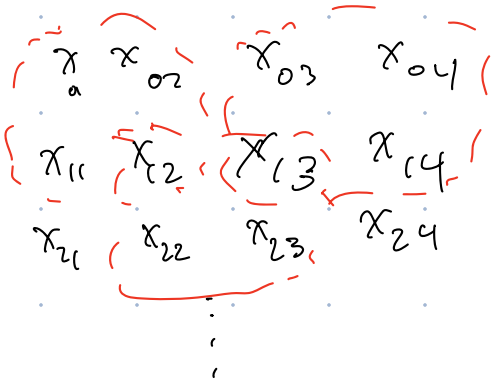
The original proof (AS, ALMSS) uses a different approach based on error correction. For a long time, nobody knew how to "quantize" these ideas, but recently that's changed.

Cook-Levin & Fault tolerance

Consider the CL tableau for a circuit



$\Rightarrow$



We know that a satisfying assignment $\approx$ a computational history of the circuit.

Turns out: an assignment violating ε-fraction of constraints $=$ a history w/ ε-fraction of faulty steps!

Suppose we had a way to make an NP-verifier fault-tolerant (so it computes the right answer even if ε-fraction of operations are faulty). Then CL on this circuit would yield a classical PCP of some kind!

(ALMSS: a constant-depth FT verifier for NP proofs using the Hadamard code yields a PCP w/ exp-long proof

ABGNV'06: a log-depth verifier + new FT scheme $\Rightarrow$ PCP w/ $1/\text{polylog}$ soundness)

FK vs CL:

Unfortunately, it is totally false that a state w/ $\langle\psi|H_{FK}|\psi\rangle \leq \varepsilon \cdot m \Rightarrow$ a history w/ ε-fraction faults.

E.g. $|\psi\rangle = |0^n\rangle_{\text{data}} \otimes |0\rangle_{\text{clock}}$

vs.

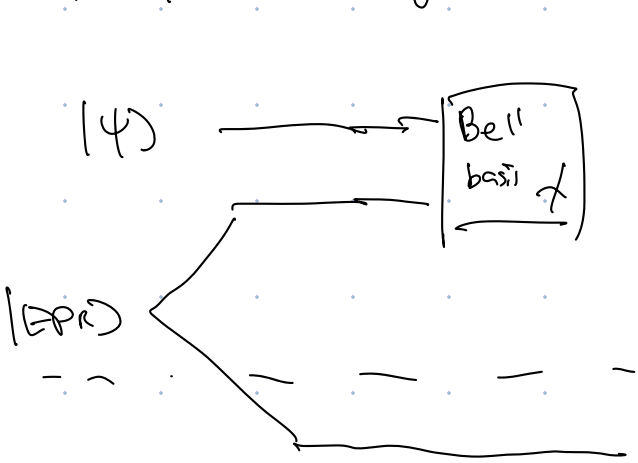
$$|\psi_{\text{history}}\rangle = \frac{1}{\sqrt{T+1}} \sum_{t=0}^{T} U_t \cdots U_1 |0^n\rangle \otimes |t\rangle$$

$|\psi\rangle$ only violates one term: the propagation term for $t=0 \Rightarrow t=1$!

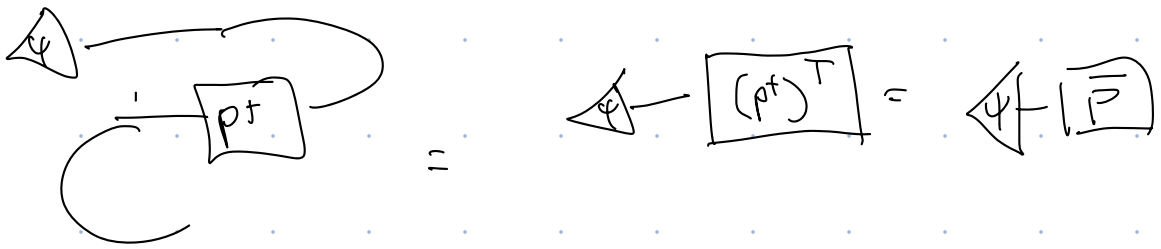
So to implement the FT → qPCP idea, need a new way to map circuits to Hamiltonians.

An interlude: teleportation

Recall the quantum circuit for teleportation



If the Bell basis measurement yields outcome P, then the resulting state is $\overline{P}|\psi\rangle = P^*|\psi\rangle$ ← complex conjugate

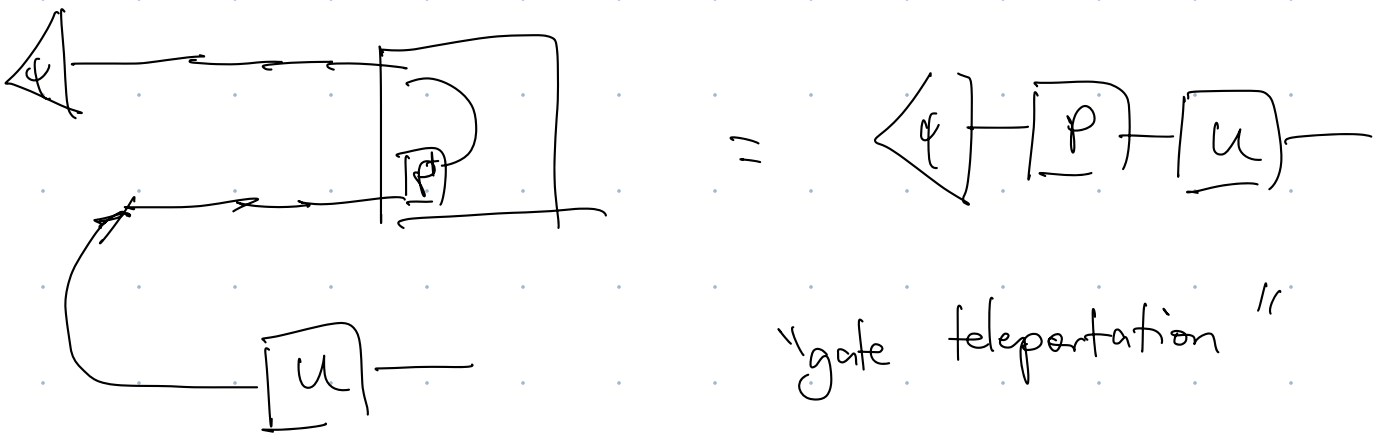


Here we have introduced the tensor diagram notation: a tensor (vector/matrix/multilinear operator) has a leg for each index, and legs are

oriented (a right leg = row index, left leg = col. index)

Normally in teleportation, P is communicated classically
→ "correction" applied by receiver

We could try sticking in a gate:



"gate teleportation"

A new circuit to Hamiltonian

This construction is due to Anshu, Breuckmann, Nguyen '23, building on some earlier papers

Motivation:

- Inspired by Detectability Lemma + CL, try to design a sequence of constraints such that applying all of them forces state to be a computational history => $|\psi_{out}\rangle$ of circuit is a ground state

- Perform gates using gate teleportation.

Consider a circuit with I gate U on I qubit. Prepare the state

$$|\psi_{in}\rangle_1 \; |\Phi^+\rangle_{23}$$

$$(I \otimes U)\,|EPR\rangle$$

Now, if we project 12 on the EPR state, the result is

$$\alpha \; |EPR\rangle\, U|\psi\rangle$$

and if we project on $|\Phi_p\rangle$, result is

$$|\Phi_p\rangle\, UP|\psi\rangle.$$

What about more gates?

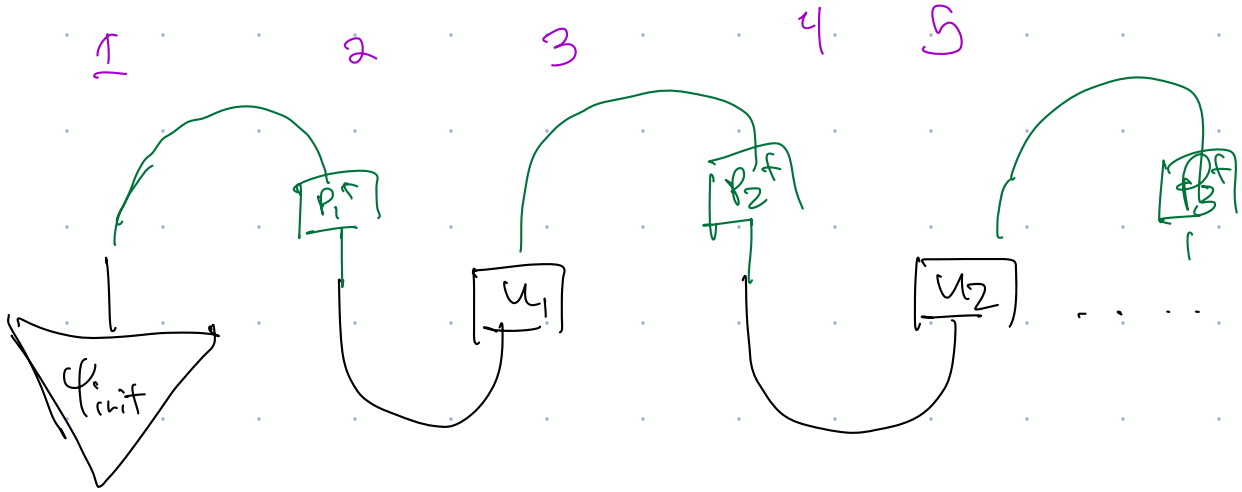
$$|\psi_{init}\rangle |\Phi_{u_1}\rangle |\Phi_{u_2}\rangle \cdots |\Phi_{u_T}\rangle$$

1 | 23 | 45 | $2T, 2T+1$

could be many qubits

Project each pair $(1,2)$ $(3,4)$ $\cdots$ $(2T-1, 2T)$

onto $|\Phi_{p_1}\rangle$ $|\Phi_{p_2}\rangle$ $\cdots$ $|\Phi_{p}\rangle$



$$= U_T P_T U_{T-1} P_{T-1} \cdots U_1 P_1 |\psi\rangle$$

So applying all these projections in sequence force the initial state to become the output.

(if $P_1 = \cdots = P_T = I$)

How do we get a local Hamiltonian out?

For simplicity, consider computations starting at $|0^n\rangle$. Then the state

$$|\psi_0\rangle = |0^n\rangle |\Phi_{u_1}\rangle \cdots |\Phi_{u_t}\rangle$$

is the unique g.s. of a local Ham:

$$H_0 = \sum_{j=1}^{n} |1\rangle\langle 1|_j + (I - \Phi_{u_1})_{23} + (I - \Phi_{u_2})_{45} + \cdots$$

What we actually want is $\Pi |\psi_0\rangle$ to be our g.s.

(Π: projector)

If we could do $H = \Pi^{-1} H_0 \Pi^{-1}$, then we'd have it maybe?

Claim: If H_0 has unique g.s. $|\psi_0\rangle$, and M is invertible, then $\underbrace{M^{-1} H_0 M^{-1}}_{H}$ has unique g.s. $M|\psi_0\rangle$

Pf:

$M^{-1} H_0 M^{-1} M |\psi_0\rangle = M^{-1} H_0 |\psi_0\rangle = 0$ ✓

$M^{-1} H_0 M^{-1} |v\rangle = 0 \Rightarrow H_0 (M^{-1}|v\rangle) = 0 \Rightarrow |v\rangle \propto M_0 |\psi\rangle$

Solution: perturb Π by a bit so that it becomes invertible!

on a single gate

$$\Pi = |\Phi_I\rangle\langle\Phi_I|$$

$$Q_\delta := |\Phi_I\rangle\langle\Phi_I| + \delta \sum_{P \in \{X,Z,\mathbb{I}\}^{\otimes 2} \setminus \mathbb{I}} |\Phi_P\rangle\langle\Phi_P| \approx \Pi \quad O(\delta)$$

$$\Lambda_\delta := \delta\,|\Phi_I\rangle\langle\Phi_I| + \sum_{P \in \{X,Z,\mathbb{I}\}^{\otimes 2} \setminus \mathbb{I}} |\Phi_P\rangle\langle\Phi_P|$$

$$\Lambda_\delta Q_\delta = \delta \cdot \mathbb{I} \Rightarrow \Lambda_\delta = \delta Q_\delta^{-1}$$

Define this way so $\|\Lambda_\delta\| = 1$, not $1/\delta \gg 1$.

Our final Hamiltonian will be

$$H = \Lambda_\delta^{\otimes T} (H_0) \Lambda_\delta^{\otimes T}$$

Not hard to check that the unique g.s. of H is

$$|\psi_\delta\rangle \propto \sum_{\vec{p}} \delta^{|\vec{p}|} |\Phi_{\vec{p}}\rangle \otimes \underbrace{U_T P_T \cdots U_1 P_1 |\psi\rangle}_{\text{history with faults } \vec{p}}$$

$\vec{p}$: vector of T Paulis

The weight $\delta|\bar{P}|$ means we have δ fraction of faults on average.

Low-energy states of H

Ultimately our goal was to find an H s.t. low-energy states correspond to executions with few faults.

In fact, for our H, even the g.s. has some stochastic faults (at a rate of δ per gate)

Conjecture: Any state of H w/ energy $\varepsilon \cdot m$ $\hat{\wedge}$ output of computation on a computation w/ ε + a little bit fraction adversarially chosen faults
(= could be anywhere)

ABN prove this for "combinatorial" states: $|\psi\rangle$ that violate $\varepsilon \cdot m$ fraction of terms.

Energy $\varepsilon \cdot m$: $\langle\psi| H |\psi\rangle = \varepsilon \cdot m$

Combinatorial error $\varepsilon \cdot m$: $\left|\{i : h_i|\psi\rangle \neq 0\}\right| = \varepsilon \cdot m$

Intuition for how this avoids local indistinguishability

Recall that we motivated FK by the problem of certifying the CAT^+ state $\Rightarrow$ CAT^- state

$|CAT^+\rangle$ $= |CAT^-\rangle$

How does ABN handle this? The "propagation Hamiltonian" is

$$h_{prop,2} = \Lambda_\delta^{\otimes n}\Big(\underbrace{(I - \Phi_2)_1}_{h_{prop,0}} + \sum_j (I - \Phi_I)_j \Big) \Lambda_\delta^{\otimes n}$$

$$gs(h_{prop,2}) = \{ Q_\delta^{\otimes n} |\psi\rangle : h_{prop,0} |\psi\rangle = 0 \}$$

$$= \{ Q_\delta^{\otimes n} |\psi_{in}\rangle |\Phi_2\rangle |\Phi_I\rangle \dots |\Phi_I\rangle \}$$

$$= \{ \sum_p \delta^{|p|} |\Phi_p\rangle_{2,I} P |\psi_{in}\rangle \}$$

Compare this to the ground space of the propagation of the identity gate

$$gs(h_{prop,I}) = \{ \sum_p \delta^{|p|} |\Phi_p\rangle_{1,I} P |\psi_{in}\rangle \}$$

We can commute the Z and the P:

$$\left(\sum_p (-1)^{\text{sgn}(p,z)} \delta^{|\hat{p}|} |\Phi_{\vec{p}}\rangle\langle p| Z |\psi_{in}\rangle\right) \quad \text{vs} \quad \left(\sum_p \delta^{|\hat{p}|} |\Phi_{\vec{p}}\rangle\langle p| |\psi_{in}\rangle\right)$$

locally can detect this relative phase if $\delta > 0$

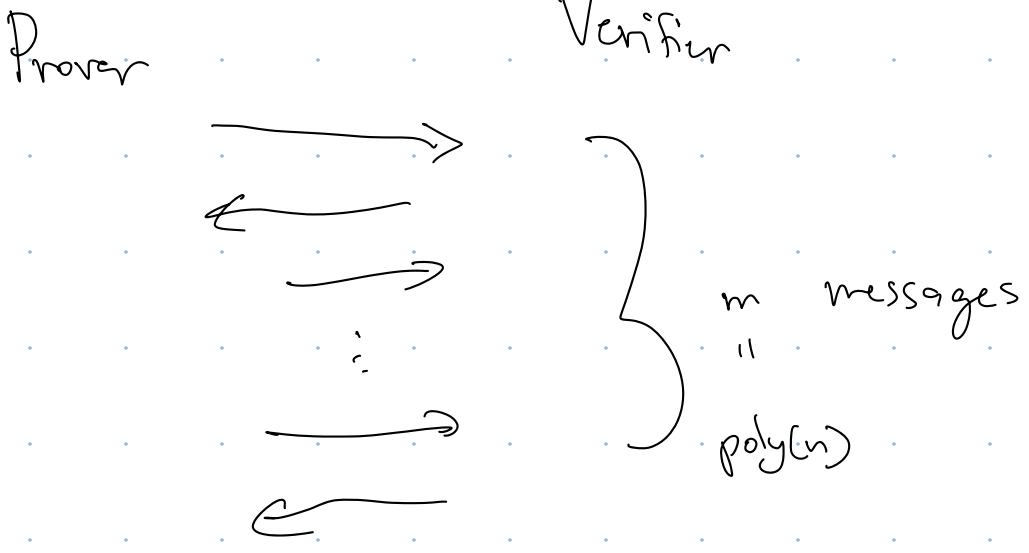
(11/5/26):

Lecture 16: QIP

So far we've talked a lot about QMA & friends

Merlin $\xrightarrow{1\text{D}}$ Arthur

This is a non-interactive proof system. We're now going to see what you can get if you add the power of interaction.



Classically, this class is called IP.

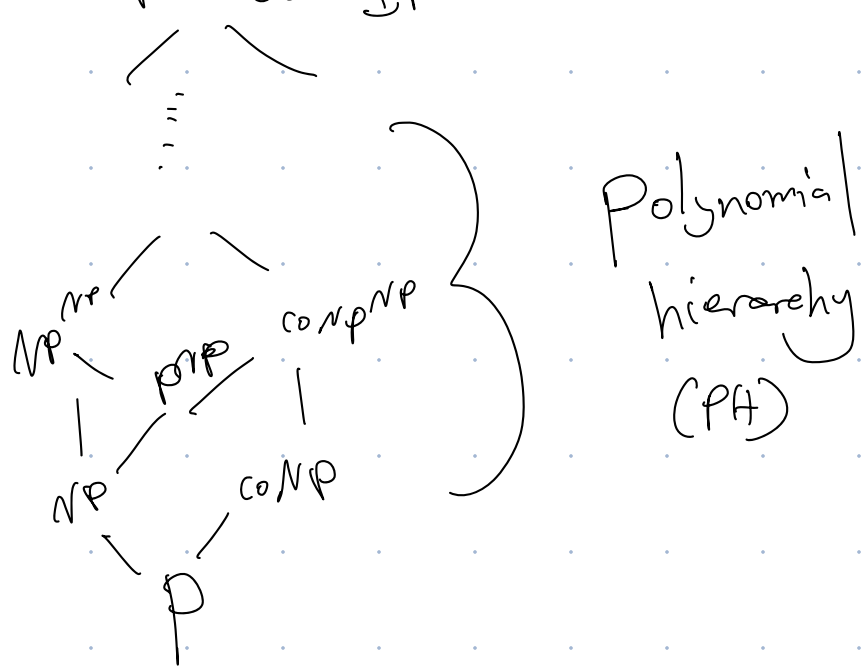
Obviously $NP \subseteq IP$.

Slightly less obviously, $IP \subseteq PSPACE$

Very non-obviously, $IP = PSPACE$!

$\Rightarrow$ interaction adds a ton of power!

In fact, we have a whole landscape:



The PH is an infinite tower of complexity classes that are all believed to be distinct.

It turns out this implies that IP requires poly-many rounds:

IP with $m = O(1)$ messages $\in AM \in coNP^{NP}$

↑ two messages

The quantum version of this class is

QIP = interactive proofs ~/
quantum prover & verifier
quantum messages
poly(n) messages / qubits of communication

It's clear that $IP \subseteq QIP$

↑ just measure the messages from the prover

But what else can it do?

Ex: Quantum circuit distinguishability

Given two circuits C_1, C_2 implementing quantum channels Φ_1, Φ_2, decide

whether $\exists\ \rho_{AB}$ s.t.

$$\left\| (\Phi_{1,A} \otimes I_B)(\rho_{AB}) - (\Phi_{2,1} \otimes I_B)(\rho_{AB}) \right\|_1 \geq \alpha$$

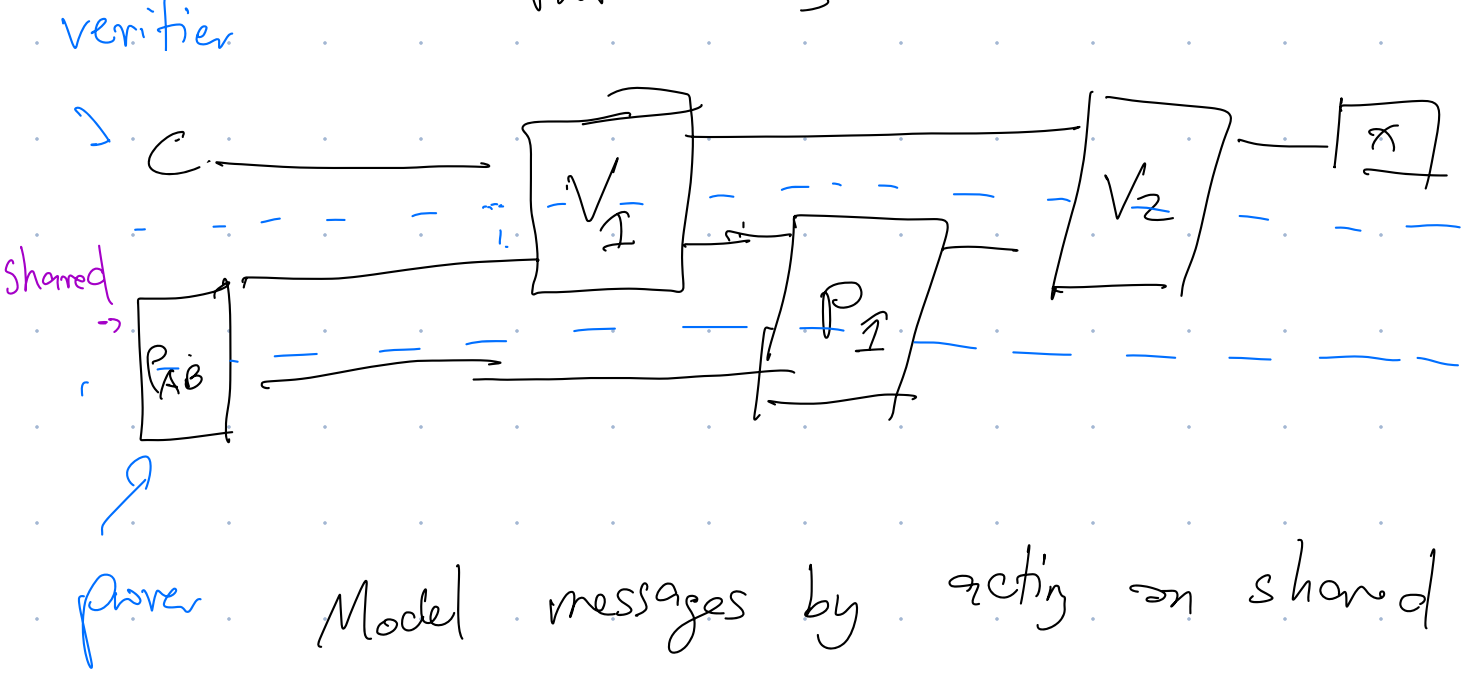
or $\forall\ \rho_{AB}$, $\| \cdots \| \leq \beta$

Protocol: This protocol has 3 messages.

Prover $\xrightarrow{1}$ Verifier

$\xrightarrow{2}$

Prover $\xrightarrow{3}$ Verifier



Model messages by acting on shared registers.

Protocol: 1) Prover prepares distinguishing state ρ_{AB}, sends B to Verifier

2) Verifier privately flips coin $b \in \{0,1\}$ to choose ϕ_b and applies it to B. Sends back to prover

3) Prover guesses b', returns to verifier

4) Verifier accepts iff $b = b'$.

Round reduction for QIP:

We said earlier that IP needs poly-many rounds. It turns out that quantumly, this is not true:

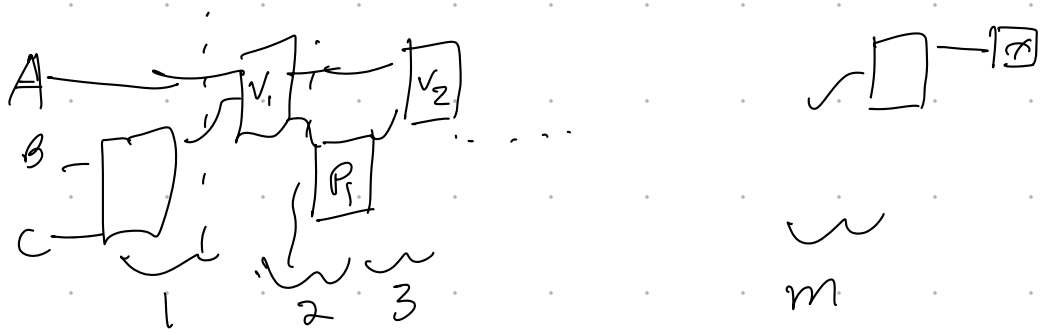
QIP = QIP(3 messages)

The proof of this is by a "cut and choose" technique.

We iteratively reduce the number of messages

$$QIP(m)_{c=1,\ s=1-\varepsilon} \Longrightarrow QIP\left(\frac{m+1}{2}\right)_{c=1,\ s=1-\varepsilon/4}$$

assume odd



Let's purify the actions of prover & verifier, so their operations are all unitaries.

New protocol:

1) Prover sends state $|\psi\rangle_{ABC}$ ostensibly = state of old protocol after $\frac{m+1}{2}$ messages.

2) Verifier flips a coin $b \in \{0,1\}$

if $b=0$

- run protocol forward until message m perform original acceptance measurement

if $b=1$

- run protocol backwards until message 1.

Check that verifier's register A is returned to $|0^\ell\rangle$

This is a unique quantum step. Classically, impossible to "unflip" the verifier's random coins!

Completeness is obvious

Soundness requires a useful technical tool

Uhlmann's theorem

Recall the Fidelity

$$F(\rho,\sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1$$

$(=1$ if $\rho=\sigma$, $=0$ if orthogonal$)$

Then $F(\rho,\sigma) = \max_{|\psi\rangle_{AB}, |\phi_{AB}\rangle} |\langle\psi|\phi\rangle|$

s.t.

$\mathrm{tr}_B |\psi\rangle\langle\psi| = \rho$

$\mathrm{tr}_B |\phi\rangle\langle\phi| = \sigma$

Related, $\mathrm{tr}_B |\psi\rangle\langle\psi| = \mathrm{tr}_B |\phi\rangle\langle\phi| = \rho$ iff

$$|\phi\rangle\langle\phi| = I_A \otimes U_B |\psi\rangle\langle\psi|$$

The fidelity is related to the trace distance

$$1 - F(\rho,\sigma) \leq \frac{1}{2}\|\rho-\sigma\|_1 \leq \sqrt{1-F(\rho,\sigma)^2}$$

Let's use this to analyze the soundness of the round-reduced protocol.

Let $\mathcal{C}_{\text{forward}} = \{ \mathrm{tr}_{B_{t-1}C_{t-1}}[\psi] : |\psi\rangle_{A_{t-1},B_{t-1},C_{t-1}}$ original obtained by running the protocol forwards $\}$

$\mathcal{C}_{\text{backward}} = \{ \mathrm{tr}_{B_{t-1},C_{t-1}}[\psi] : |\psi\rangle_{A_{t-1},B_{t-1},C_{t-1}}$ obtained by running protocol backwards from accepting $\}$

by Uhlmann

$$\text{Coc}(\text{protocol}) \overset{?}{=} \max_{\sigma \in \mathcal{C}_{\text{forward}}} \max_{\tau \in \mathcal{C}_{\text{backward}}} F(\sigma,\tau)^2 \leq 1-\varepsilon$$

↑ max prob. of success

↑ by assumption

$$\omega(\text{new protocol}) = \max_{\substack{\sigma \in \mathcal{C}_{\text{forward}} \\ \tau \in \mathcal{C}_{\text{backward}}}} \frac{F(\rho,\sigma)^2 + F(\rho,\tau)^2}{2}$$

ρ

by a fidelity fact $\rightarrow$

$$= \max_{\substack{\sigma \in \mathcal{C}_{\text{forward}} \\ \tau \in \mathcal{C}_{\text{backward}}}} \frac{1 + F(\sigma,\tau)}{2}$$

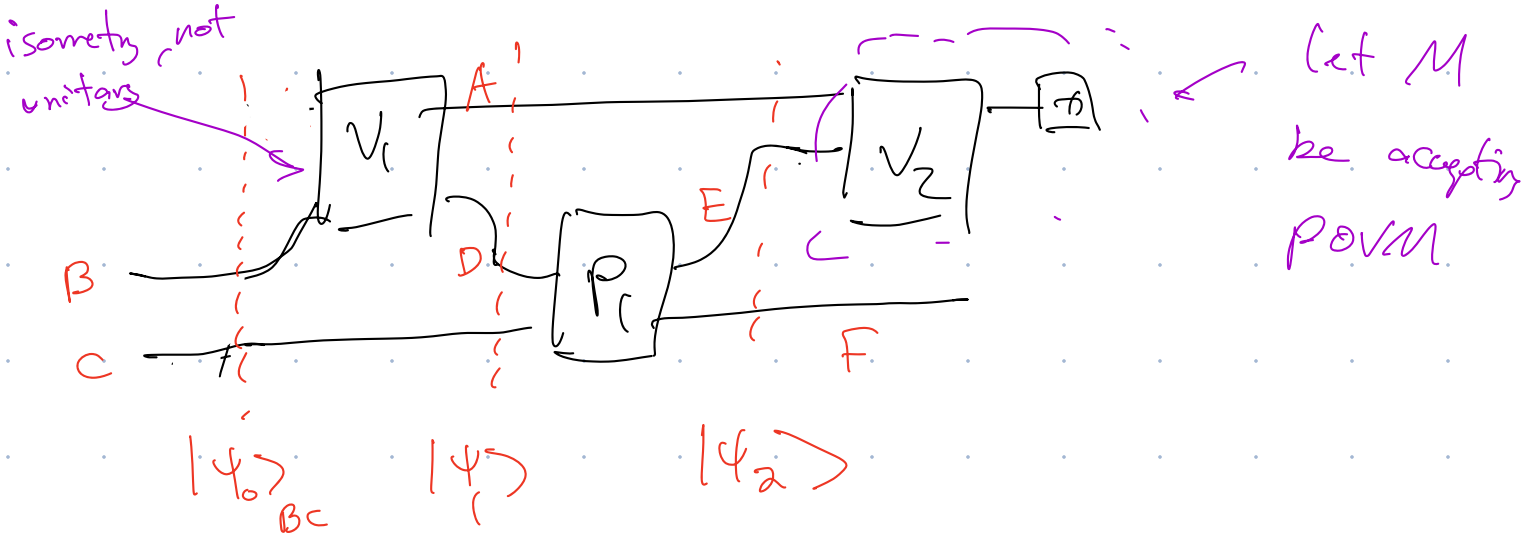
$$\leq \frac{1}{2} + \frac{1}{2}\sqrt{1-\varepsilon} \leq 1 - \frac{\varepsilon}{4}$$

In general, for an m-round protocol, we'll have to repeat this $\log_2 m$ times

sounds $1-\varepsilon \Rightarrow 1 - \varepsilon/4^{\log_2 m} = 1 - \frac{\varepsilon}{m^2}$

Can we boost this back up?

SDPs & parallel repetition



$$\omega(V) = \max \operatorname{tr}\left[(M_{AE} \otimes I_F)\, \psi_2 \right]$$

s.t. $\exists\, P_i, |\psi_1\rangle, |\psi_0\rangle$

$$|\psi_2\rangle = (I_A \otimes P_{i\,BC})\, |\psi_1\rangle$$

$$|\psi_1\rangle = V_1 |\psi_0\rangle$$

Uhlmann

$$\max_{\rho_{AE},\, \sigma} \operatorname{tr}[M \rho]$$

s.t.

$$\rho_A = \operatorname{tr}_{BC}\left[V_1\, \sigma_{BC}\, V_1^{\dagger} \right]$$

$$\rho \geq 0 \qquad \operatorname{tr}\rho = 1$$

$$\sigma \geq 0 \qquad \operatorname{tr}\sigma = 1$$

This is a semidefinite program!

It has a dual

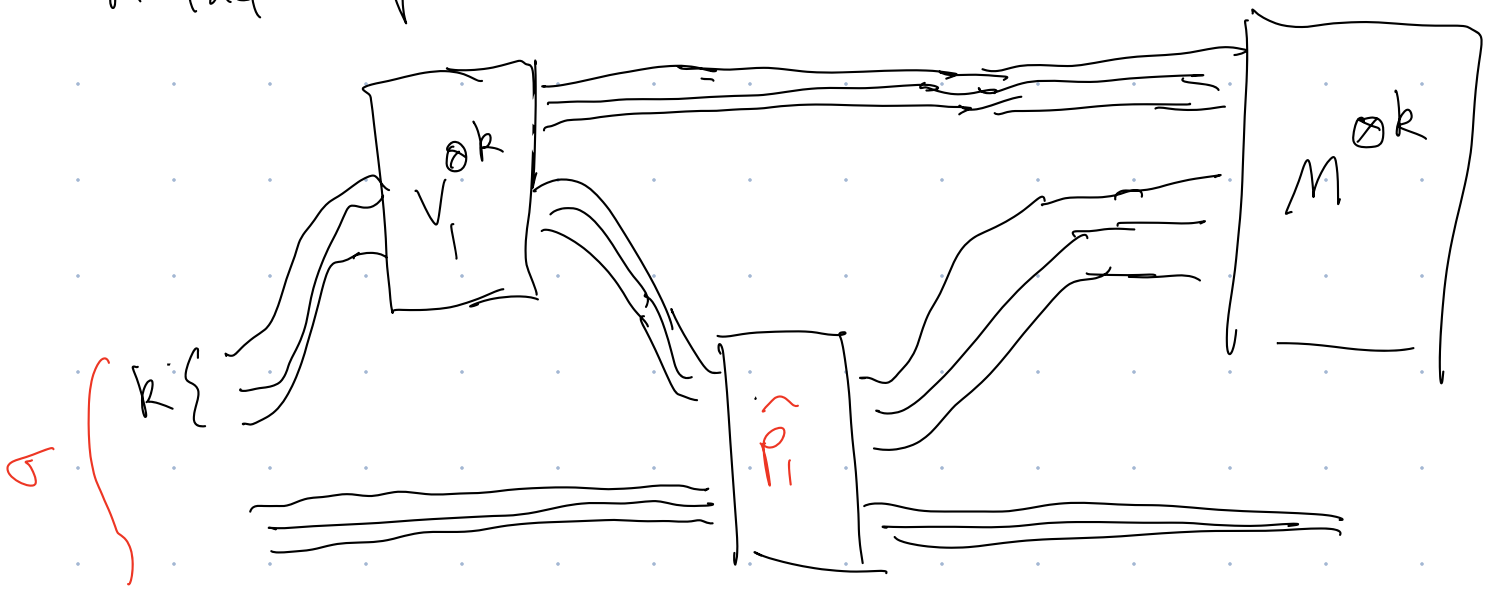
$$\begin{aligned} \min_{Y,\lambda} \quad & \lambda \\ \text{s.t.} \quad & Y_A \otimes I_E \geq M_{AE} \\ & V_1^\dagger (Y_A \otimes I_D) V_1 \leq \lambda I_B \\ & Y = Y^\dagger \end{aligned}$$

The idea is that feasible dual points certify upper bounds on the primal

$$\begin{aligned} \operatorname{tr} M\rho \leq \operatorname{tr}\left[(Y \otimes I)\rho_{AE}\right] &= \operatorname{tr}[Y\rho_A] \\ &= \operatorname{tr}\left[(Y \otimes I) V_1 \sigma_{BC} V_1^\dagger\right] \\ &= \operatorname{tr}\left[V_1^\dagger (Y_A \otimes I_{DC}) V_1 \sigma_{BC}\right] \\ &\leq \lambda \end{aligned}$$

This implies perfect parallel repetition!

k-fold repetition:



Note: $\tilde{P}_1$ is a prover operation. No guarantee $\tilde{P}_1 = P_1^{\otimes k}$, or that initial state is a tensor product

Tensor product strat

$$\Rightarrow \omega\left(k\text{-fold repetition}\right) \geq \omega(V)^k$$

Tensor dual certificate

$$\Rightarrow \omega\left(k\text{-fold repetition}\right) \leq \omega(V)^k$$

pf:

$\omega(V) = \alpha \Rightarrow \exists\ \lambda, y$ achieving objective value α on dual

↑ strong duality

$\Downarrow$

$\lambda^k, y^{\otimes k}$ achieve value α^k on dual for k-fold repetition

$\omega(\text{$k$-fold repetition}) \le \alpha^k$ ⟸ weak duality

This SDP characterization has other uses: it shows that

$QIP \subseteq PSPACE$ [Jain, Ji, Upadhyay, Watrous]

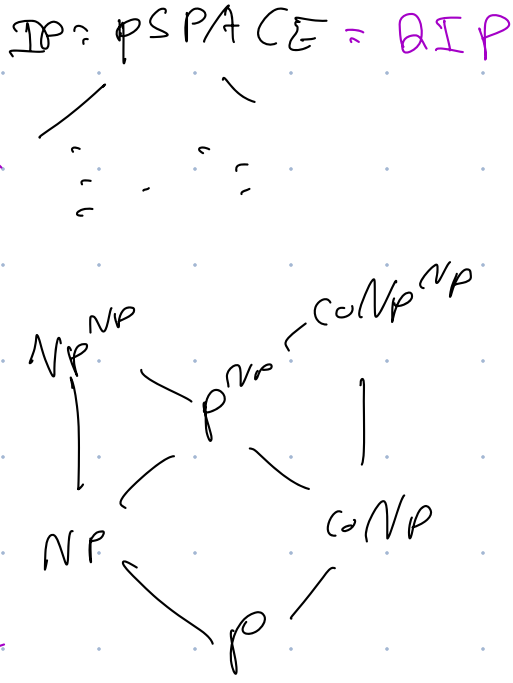
$\Rightarrow QIP = PSPACE$ ($PSPACE \subseteq QIP$ follows from $IP \subseteq QIP$)

Idea: solve SDP in PSPACE using matrix multiplicative weights alg. (exploiting special structure)

11/10/26

Lecture 17: BQP vs PH

Last time we saw



PH = Polynomial Hierarchy =

A natural question to understand is where BQP sits relative to this hierarchy. We already discussed BQP vs NP: we conjecture $BQP \not\subseteq NP$ and have oracle evidence for this. People actually believe the much stronger conjecture

Conj: $BQP \not\subseteq PH$

But finding an oracle separating these classes proved quite difficult: candidate proposed by Aaronson '09

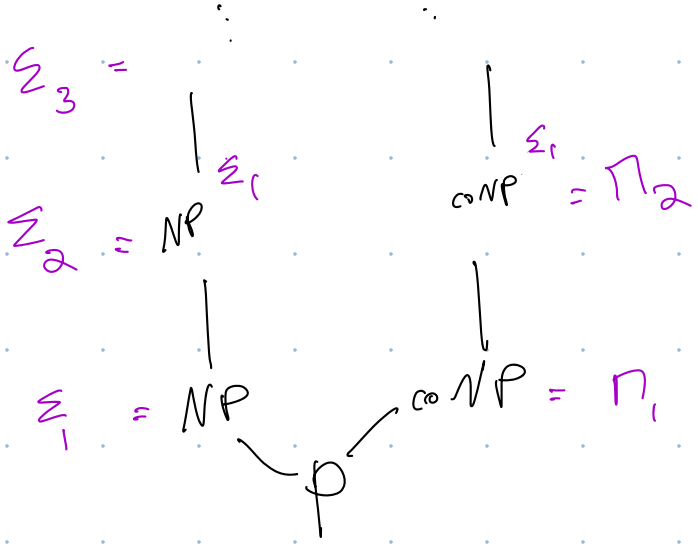
proved by Raz-Tal '18

The oracle problem that came out of this is called "Forrelation" and is very beautiful. But first, some preliminaries on PH.

"polynomial" or "protean"?

PH: quantifiers & circuits

The classes in PH have a much cleaner characterization in terms of alternating quantifiers



A language $L \in \Sigma_1$ iff $\exists$ a poly-time TM M s.t.

$$L = \{x \mid \exists w \text{ s.t. } M(x,w) = 1\}$$

(w: NP-witness)

A language $L \in \Pi_1$ (= coNP) iff $\exists$ a poly-time TM M s.t.

$$L = \{ x \mid \forall w \quad M(x, w) = 1 \}$$

It turns out that $L \in \Sigma_2$ (= NP^{NP}) iff $\exists$ a poly-time M s.t.

$$L = \{ x \mid \exists w_1 \, \forall w_2 \quad M(x, w_1, w_2) = 1 \}$$

(Idea: $L = \{ x \mid \exists w$ s.t. $A(x, w, f_1(x,w), \ldots, f_k(x,w)) = 1 \}$

$f_1, \ldots, f_k$: NP languages

$\exists w \; \exists b_1 \ldots b_k$ s.t. $\left[A(x, w, b_1, \ldots b_k) = 1 \right] \wedge \left[\begin{array}{l} f_1(x,w) = b_1 \\ \vdots \\ f_k(x,w) = b_k \end{array} \right]$

↓ Use $\exists$ quantifiers to guess witnesses for each $f_j \stackrel{?}{=} b_j$

$$\exists w \; \exists b_1 \dots b_k \; \exists w_1^{YES} \dots w_k^{YES} \; \forall w_1^{NO} \dots w_k^{NO}$$

$$\left[A(x,w,b_1 \dots b_k) = 1\right] \wedge \left[\begin{array}{l} f_1(x,w) = b_1 \\ \vdots \\ f_k(x,w) = b_k \end{array}\right] \wedge \left[\begin{array}{l} \text{if } b_1 = \text{true} \\ \quad M_1(x,w,w_1^{YES}) = 1 \\ \text{else} \\ \quad M_1(x,w,w_1^{NO}) = 1 \end{array}\right]$$

$$\wedge \dots$$

Now, combine all the existential quantifiers into one block and all the universal quantifiers into one block.

$$\exists w_1 \; \forall w_2 \; M(x,w_1,w_2) = 1 \quad)$$

In general $L \in \Sigma_k$ iff $\exists$ poly-time M s.t.

$$L = \{x \mid \exists w_1 \forall w_2 \dots \; Q_k w_k \; M(x,w_1,\dots,w_k) = 1\}$$

$\exists$ if k odd

$\forall$ if k even

$L \in \Pi_k$ iff $\exists$ poly-time M s.t.

$$L = \{x \mid \forall w_1 \exists w_2 \cdots Q w_k \; M(x, w_1, \ldots, w_k) = 1\}$$

Q = $\forall$ if k odd
$\exists$ if k even

This characterization of PH now lets us get another characteization in terms of Boolean circuits.

To simplify, let's think about the grey model, where we only count the # of greis, not the runtime.

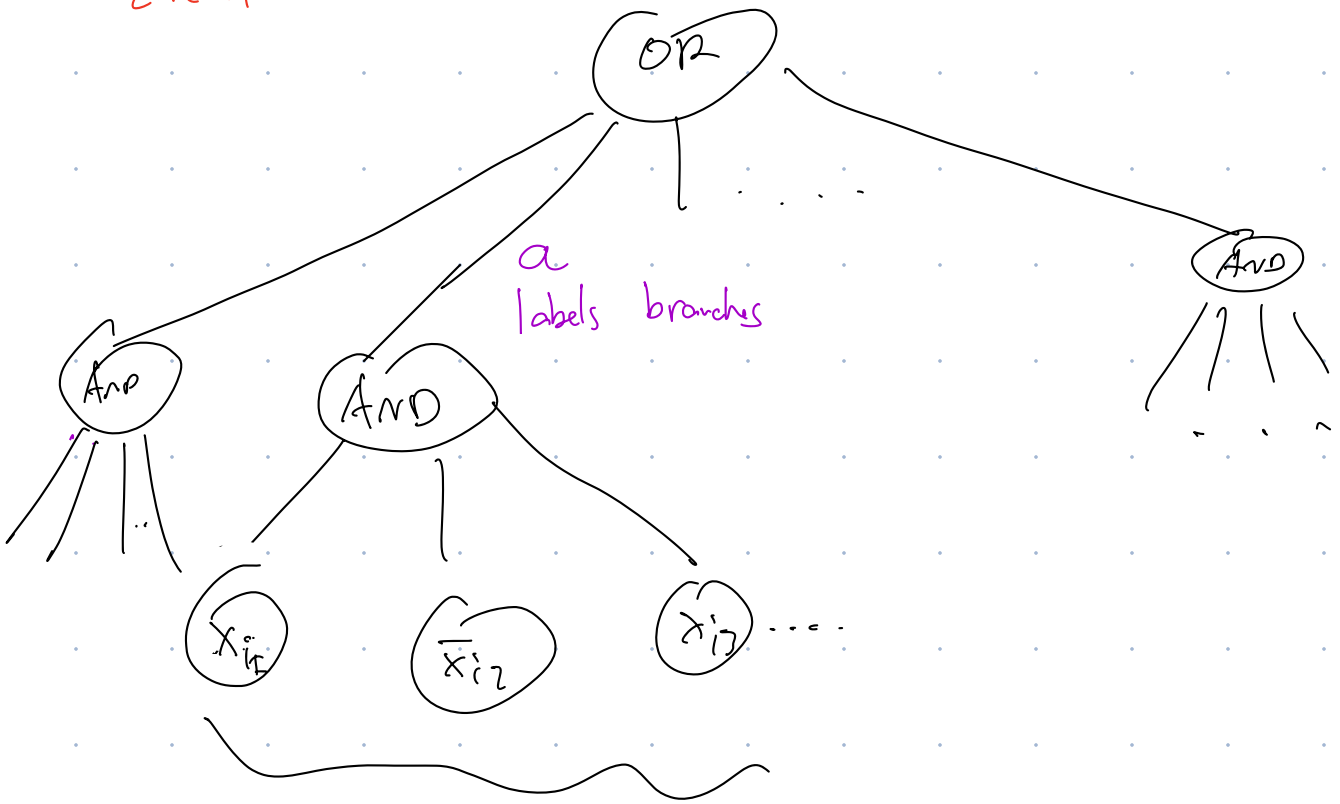
A poly-time deterministic alg. just looks up some $poly(n)$ locations in the Oracle input $x \in \{0,1\}^N$ and accepts iff they are equal to a "good" value

$$M^x = \bigvee_{(i_1 \ldots i_r, a) \in Good} \mathbb{1}\left[x_{i_1} \; x_{i_r} = a \right]$$

the locations greied by the alg. if the view it sees is $= a$

Note: you used this perspective on Pset I!

We can write this as a constant-depth, poly(n)-size circuit



Implements check

$x_{i_1} \ldots x_{i_T} = a$

(a is encoded in negation patterns)

This circuit has very special structure: it consists of AND, OR, and NOT gates, where all the NOT gates appear at the bottom layer. Such a circuit is called an AC^0 circuit

So to prove that an oracle problem is not in PH, it suffices to show that it has no AC^0 circuit.

In general, circuit lower bounds are very difficult to show, but some powerful techniques are known for AC^0. E.g. Parity $\notin AC^0$ (determining if Hamming weight of $x \in \{0,1\}^n$ is even or odd)

Back to quantum

Recall that we were trying to show that $\exists$ oracle O s.t. $BQP^O \not\subseteq PH^O$. It turns out that there is a very natural candidate that is in BQP almost by construction.

Let $f: \{0,1\}^n \to \mathbb{R}$. The Fourier transform of f is defined by $\hat{f}: \{0,1\}^n \to \mathbb{R}$

$$\hat{f}(s) = \frac{1}{\sqrt{2^n}} \sum_{x \in \{\pm 1\}^n} f(x) \cdot (-1)^{x \cdot s}$$

$:= \chi_s(x)$

This is just the good old Hadamard transform.
It can be inverted:

$$f(x) = \frac{1}{\sqrt{2^n}} \sum_{S \in \{0,1\}^n} \hat{f}(S) \cdot \chi_S(x)$$

(Pf: $= \frac{1}{\sqrt{2^n}} \cdot \frac{1}{\sqrt{2^n}} \sum_{S \in \{0,1\}^n} \sum_{y \in \{0,1\}^n} f(y) \chi_S(x) \chi_S(y)$

$= \frac{1}{2^n} \cdot 2^n \cdot f(x)$ (sum over S vanishes unless $x = y$))

If we interpret S as a subset of $[n]$ and set $\chi_i = (-1)^{x_i} = (1 - 2x_i)$, we get

$$f(X) = \sum_{S \subseteq [n]} \hat{f}(S) \prod_{i \in S} X_i$$

So the Fourier coefficients $\hat{f}(S)$ are just the coefficients of each monomial when we write f as a multilinear polynomial in the variables X_i.

squared

The FORRELATION problem is the following.

Given oracle access to $f, g : \{0,1\}^n \to \{\pm 1\}$,

let $$\Phi(f,g) = \frac{1}{2^{3n/2}} \sum_{x,y \in \{0,1\}^n} f(x)\, g(y)\, (-1)^{x \cdot y}$$

$$= \frac{1}{2^n} \sum_x \hat{f}(x)\, g(x)$$

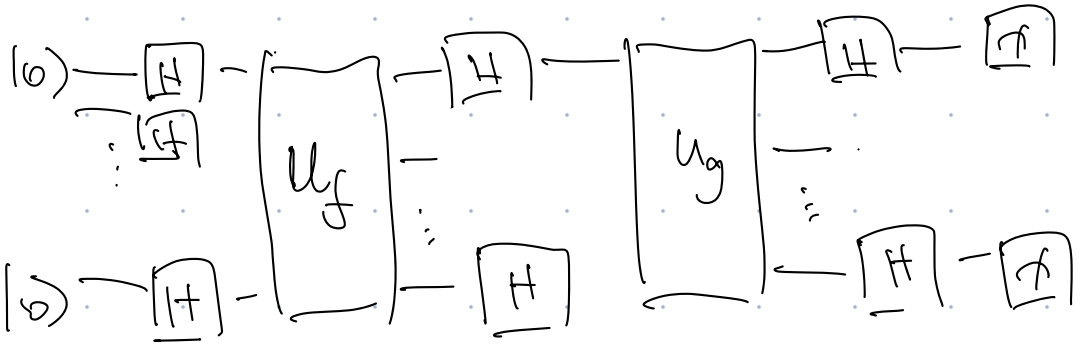
Determine whether YES: $\Phi(f,g) \geq a$

or

NO: $|\Phi(f,g)| \leq b$

separated by $1/\mathrm{poly}(n)$

This problem has a very simple quantum alg.



$$|0^n\rangle \longrightarrow \frac{1}{\sqrt{2^n}} \sum_x |x\rangle$$

$$\rightarrow \frac{1}{\sqrt{2^n}} \sum_x f(x)|x\rangle$$

$$\rightarrow \frac{1}{\sqrt{2^n}} \sum_y \underbrace{\left(\frac{1}{\sqrt{2^n}} \sum_x f(x)(-1)^{x\cdot y}\right)}_{\hat{f}(y)} |y\rangle$$

$$\rightarrow \frac{1}{\sqrt{2^n}} \sum_y g(y)\hat{f}(y)|y\rangle$$

$$\rightarrow \frac{1}{2^n} \sum_{u,y} (-1)^{u\cdot y} g(y)\hat{f}(y)|u\rangle$$

$$= \Phi(f,g)|0^n\rangle + \text{orthogonal}$$

$$\Pr[\text{measure } 0^n] = |\Phi(f,g)|^2$$

Intuitively, Forrelation requires looking at f, g in a very "global" way and so shouldn't be solvable by an AC^0 circuit $\Rightarrow$ not in PH.

Blueprint for showing Forrelation $\notin AC^0$

Consider two distributions over pairs (f, g)

1) $\mathcal{U}$: f, g both uniformly random Boolean functions chosen independently.

$$\mathop{\mathbb{E}}_{\mathcal{U}} |\Phi(f,g)|^2 = \frac{1}{2^n}$$

$$\Rightarrow \Pr\left[|\Phi(f,g)^2| > \frac{1}{\sqrt{2^n}}\right] < 1/\sqrt{2^n}$$

by Markov's ineq.

2) $\mathcal{D}$: Sample $X \in \mathbb{R}^{2^n}$ where each coord. is an indep. standard Gaussian.

$$Y := H^{\otimes n} X$$

(X, Y) $\qquad \varepsilon = \frac{1}{c \cdot n}$

$\downarrow$ rounding

(f, g)

Sample $f(x) \in \{\pm 1\}$ s.t. $\mathbb{E} f(x) = \text{truncate}(\varepsilon X_x)$

Can show that w.h.p $\Phi(f, g) \geq \text{constant} \cdot \varepsilon$

Thm [Raz-Tal]: No AC^0 circuit can distinguish D from U (D is "pseudorandom" relative to AC^0 circuits)

Why is this hard to prove?

The most basic way of showing AC^0 lower bounds is through polynomial method:

AC^0 circuit $C \longrightarrow$ A multilinear polynomial $P : \mathbb{R}^N \to \mathbb{R}$ agreeing w/ C on $\{0,1\}^N$

$N = 2^n$

on $z \in \{0,1\}^n$

↑

oracle input

P can be very high degree in general (a single OR gate gives degree N).

But it turns out that randomly fixing some coordinates z_i to 0 or 1 collapses P to a low-degree polynomial.

E.g. $OR|_{z_i = 1} = 1$ (constant polynomial)

So AC^0 can only compute functions whose "random restrictions" are low degree.

Problem: Forrelation comes from a quantum alg. so is computed by a low-degree poly!

Raz & Tal use fourier analysis (carefully controlling the high-degree coeffs. of P) + specific properties of Gaussian distribution.